

Why visualize data?

CS424: Visualization & Visual Analytics

Fabio Miranda

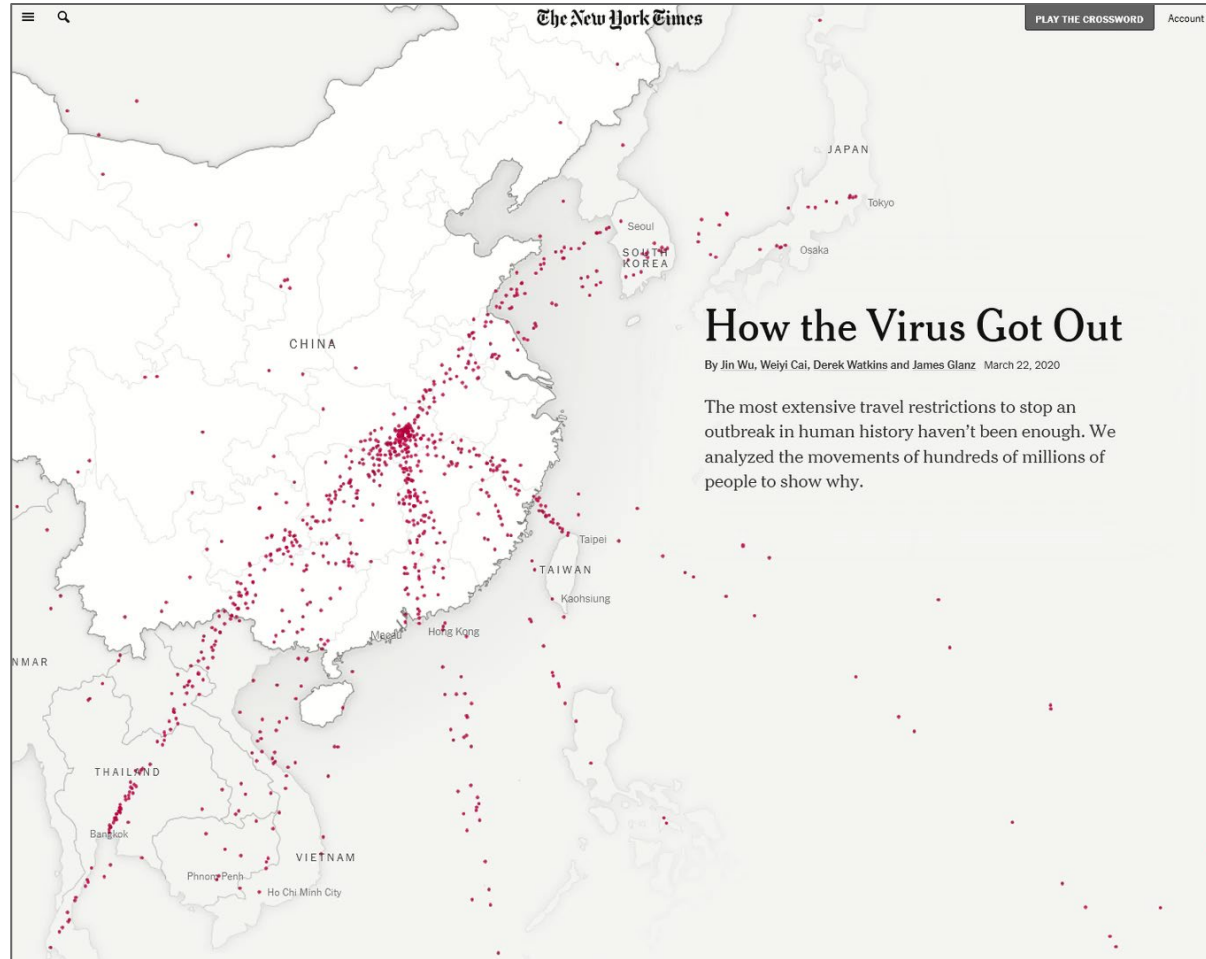
<https://fmiranda.me>

Visual analytics

“Visual analytics is the formation of abstract visual metaphors in combination with a human information interaction that enables detection of the expected and discovery of the unexpected within massive, dynamically changing information spaces.”

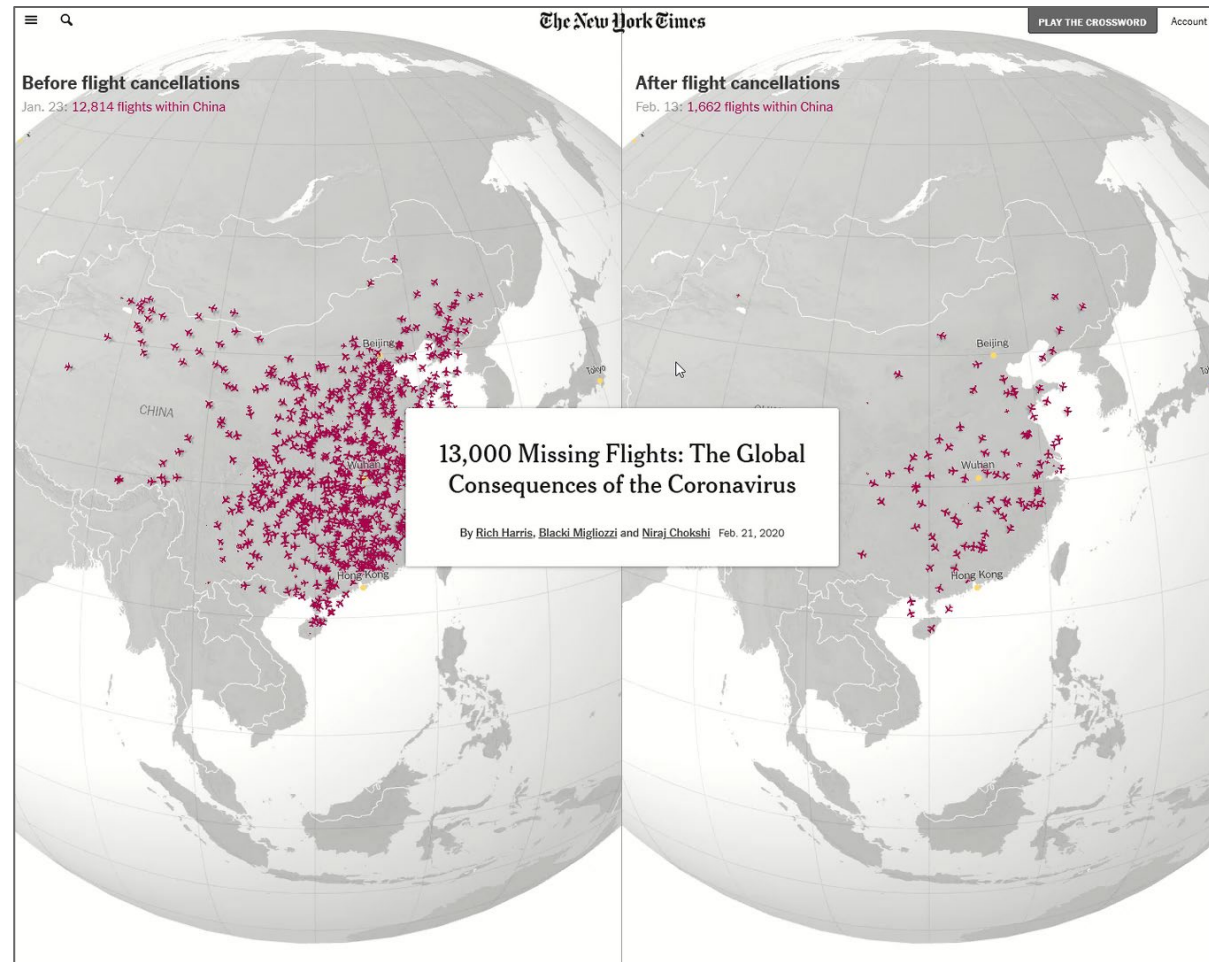
[Wong and Thomas, 2004]

Example: communicating findings



<https://www.nytimes.com/interactive/2020/03/22/world/coronavirus-spread.html>

Example: communicating findings



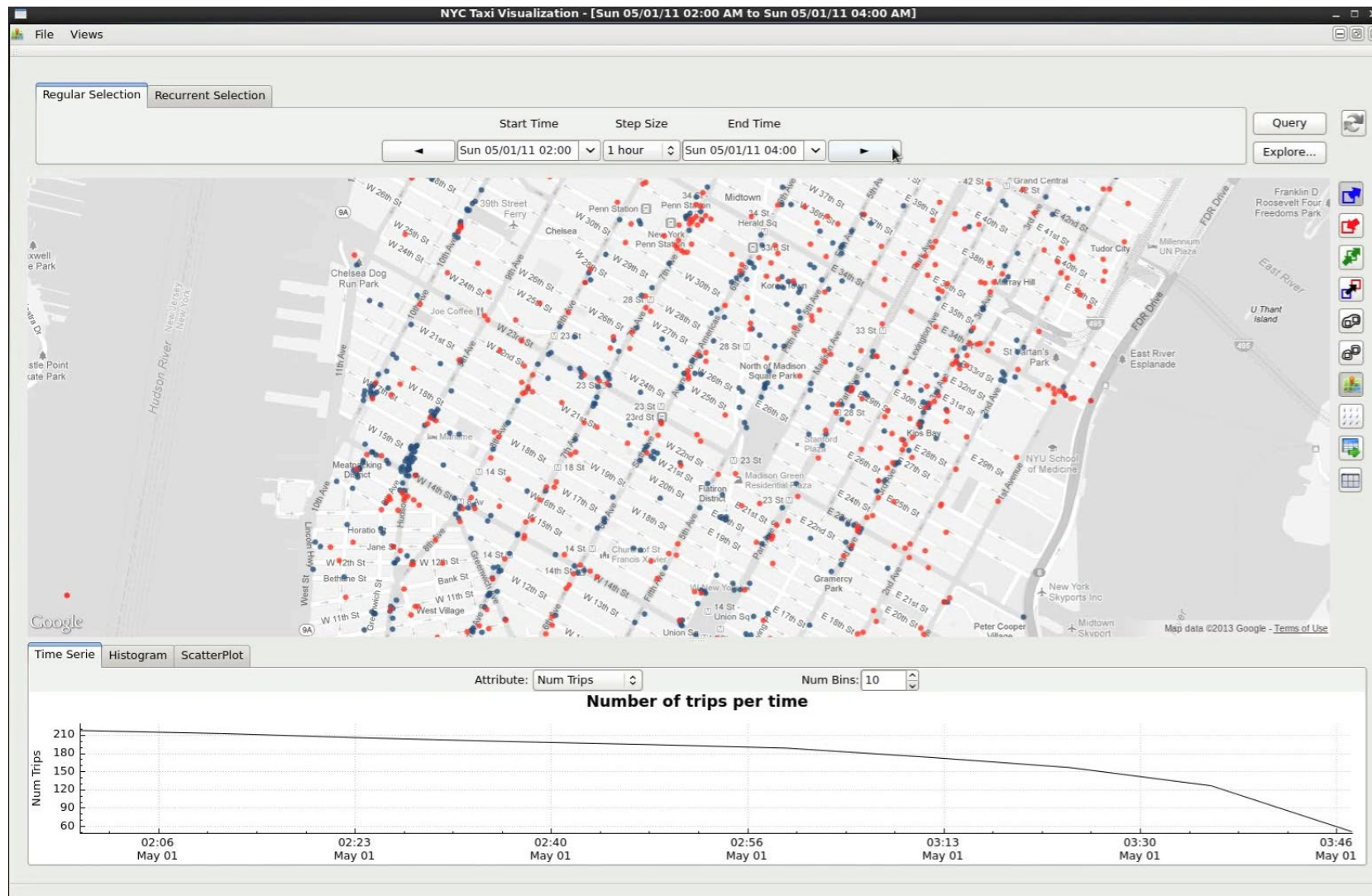
<https://www.nytimes.com/interactive/2020/02/21/business/coronavirus-airline-travel.html>

Example: exploration + communicating findings

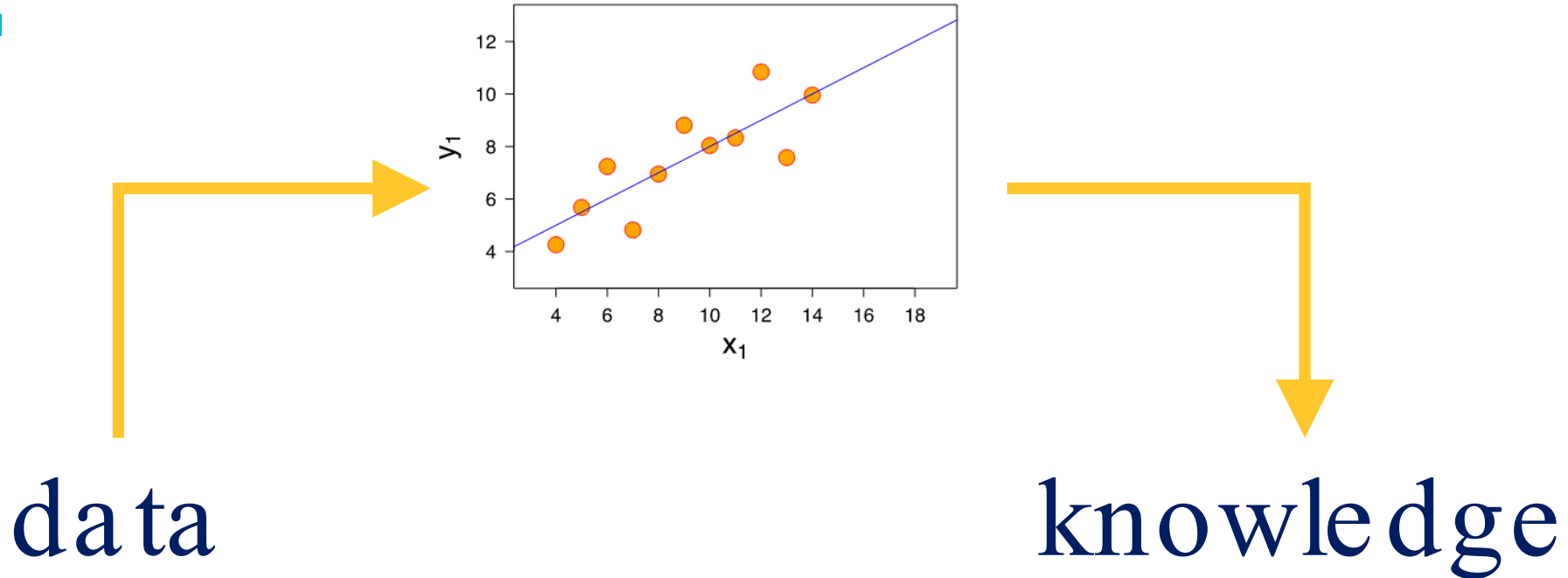


<https://www.nytimes.com/interactive/2016/12/21/upshot/Mapping-the-Shadows-of-New-York-City.html>

Example: visual analysis of data



Data to knowledge



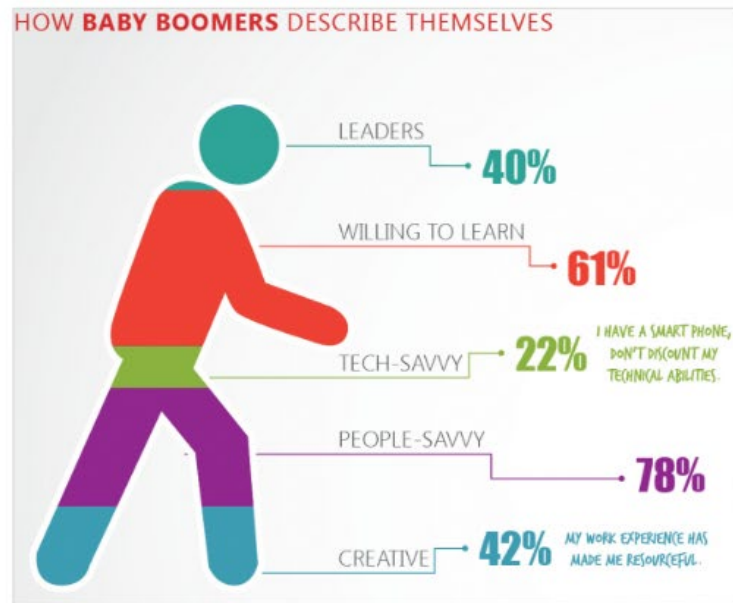
Transform data into visual marks

Data visualization



Data visualization

insight Communication → insight

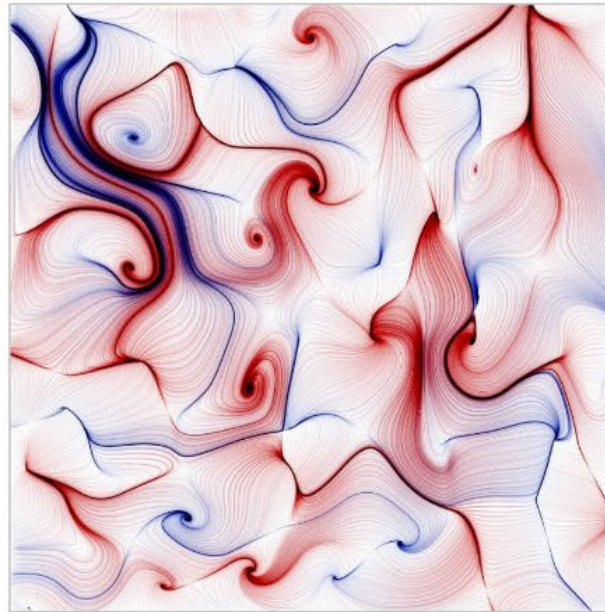
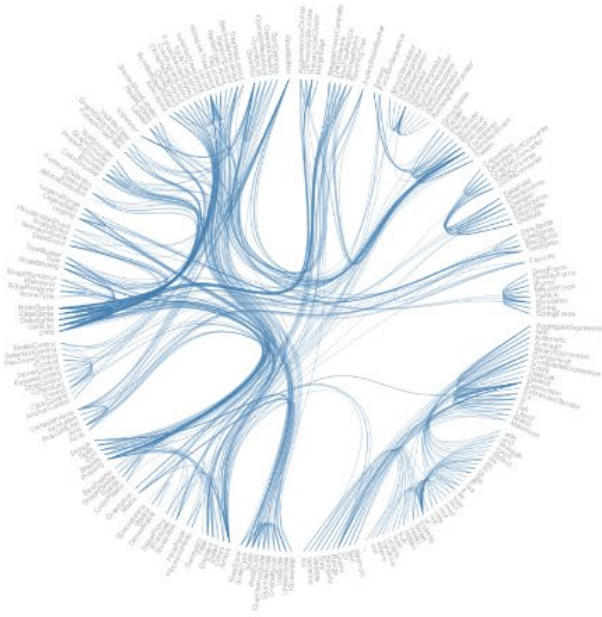


Data visualization

data

Exploration /
Analysis

insight



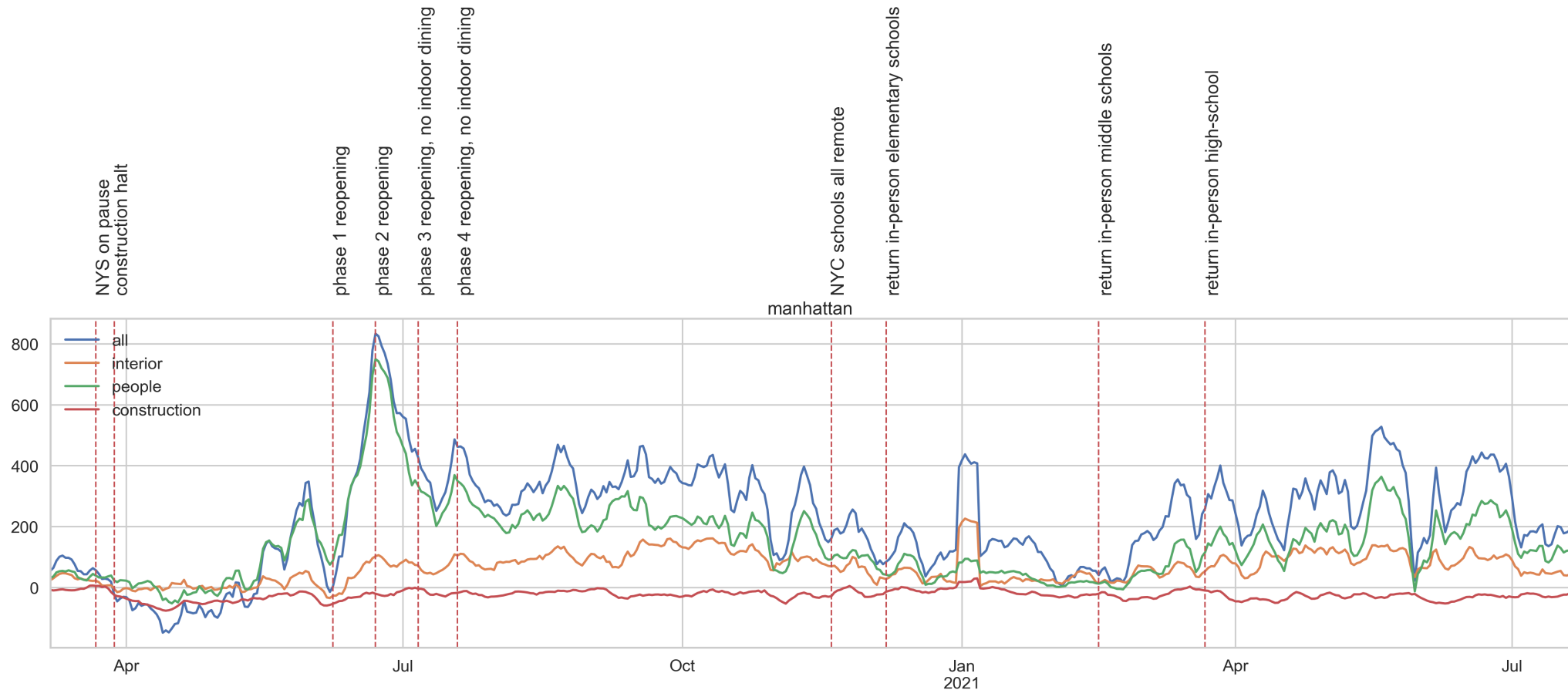
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LineSprite	RectSprite	RootScale	LogScale	
	ScaleType	IScaleMap	OrdinalScale	TimeScale
DirtySprite	TextSprite	AspectRatioBanker	SpanningTree	Betweenness
		MergeEdge	CommunityStructure	LinkDistance
IForce	Drag	GraviSpring	AdaptiveCluster	ShortestPaths
			HierarchicalCluster	MaxFlowMinCut
		Count	Sum	Minimum
			Maximum	Average
			Distinct	Or
			And	Xor
			Variable	
Spring	Particle	Literal	Not	Range
			Aggregate	Variance
			IsA	If
Simulation	NBodyForce	BinaryExpression	Fn	ExpressionFilter
				CompositeExpression
				Match
DataData	DataData	Arithmetic	Expression	Query
		-	select	variance
		count	where	xor
		sum	update	stddev
		max	orderby	not
		min	or	fn
		average	and	isa
		distinct		lte
				gte
				range

Example:

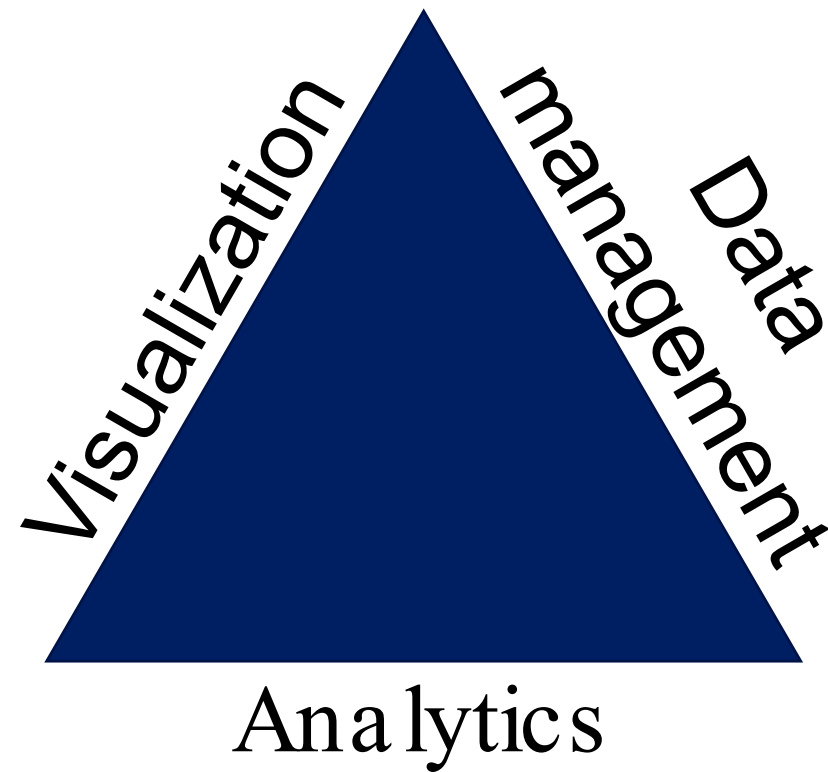
Noise complaints during pandemic

	date	unique_key	created_date	closed_date	agency	agency_name	complaint_type	descriptor	location_type	incident_zip	...	landmark	date.1	hour_of_day	week	weekday	year	day_of_month	month	aligned_day_index	datetime
0	2017-01-01	35138317	2017-01-01T00:02:54.000	2017-01-01T00:46:54.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	11209.0	...	NaN	2017-01-01 00:02:54	0	52	6	2017	1	1	0.0	2017-01-01 00:02:54
1	2017-01-01	35139300	2017-01-01T00:03:41.000	2017-01-01T03:49:13.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	10040.0	...	NaN	2017-01-01 00:03:41	0	52	6	2017	1	1	0.0	2017-01-01 00:03:41
2	2017-01-01	35137537	2017-01-01T00:04:01.000	2017-01-01T00:44:40.000	NYPD	New York City Police Department	Noise - Residential	Banging/Pounding	Residential Building/House	11214.0	...	NaN	2017-01-01 00:04:01	0	52	6	2017	1	1	0.0	2017-01-01 00:04:01
3	2017-01-01	35138401	2017-01-01T00:06:04.000	2017-01-01T01:52:03.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	11691.0	...	NaN	2017-01-01 00:06:04	0	52	6	2017	1	1	0.0	2017-01-01 00:06:04
4	2017-01-01	35139201	2017-01-01T00:08:24.000	2017-01-01T06:43:42.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	10458.0	...	NaN	2017-01-01 00:08:24	0	52	6	2017	1	1	0.0	2017-01-01 00:08:24
5	2017-01-01	35140227	2017-01-01T00:09:08.000	2017-01-01T02:16:21.000	NYPD	New York City Police Department	Noise - Residential	Loud Television	Residential Building/House	11366.0	...	NaN	2017-01-01 00:09:08	0	52	6	2017	1	1	0.0	2017-01-01 00:09:08
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7	2017-01-01	35141927	2017-01-01T00:12:02.000	2017-01-01T00:59:53.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	11204.0	...	NaN	2017-01-01 00:12:02	0	52	6	2017	1	1	0.0	2017-01-01 00:12:02
8	2017-01-01	35138731	2017-01-01T00:12:36.000	2017-01-01T08:29:48.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	10457.0	...	NaN	2017-01-01 00:12:36	0	52	6	2017	1	1	0.0	2017-01-01 00:12:36
9	2017-01-01	35141039	2017-01-01T00:12:44.000	2017-01-01T00:45:47.000	NYPD	New York City Police Department	Noise - Residential	Loud Music/Party	Residential Building/House	10312.0	...	NaN	2017-01-01 00:12:44	0	52	6	2017	1	1	0.0	2017-01-01 00:12:44

Example: Noise complaints during pandemic



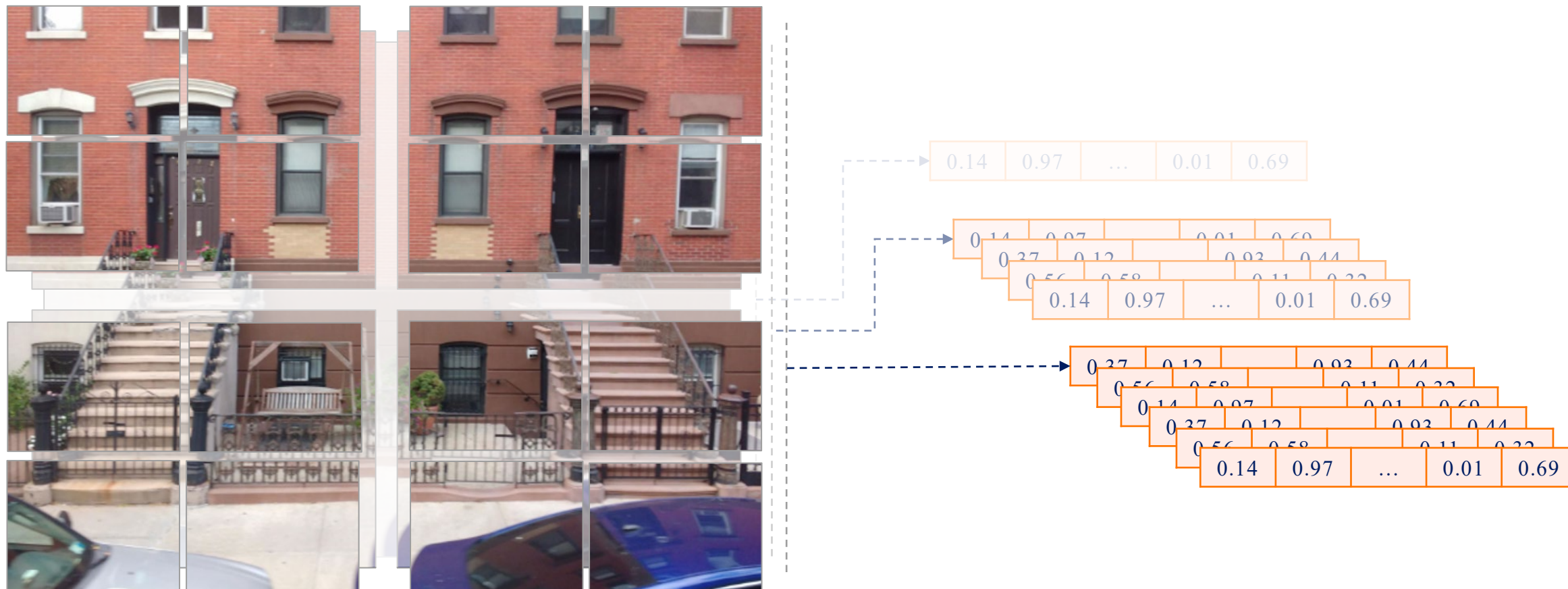
Visual analytics



Images



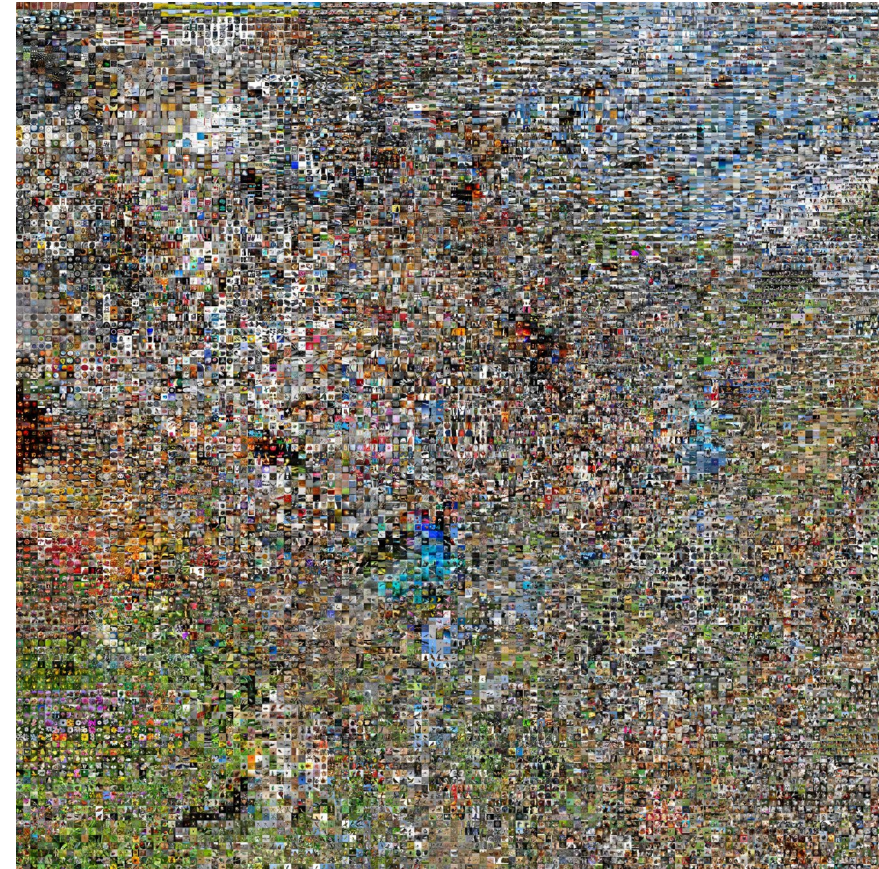
Images



tSNE



tSNE



Audio



Taxi patterns

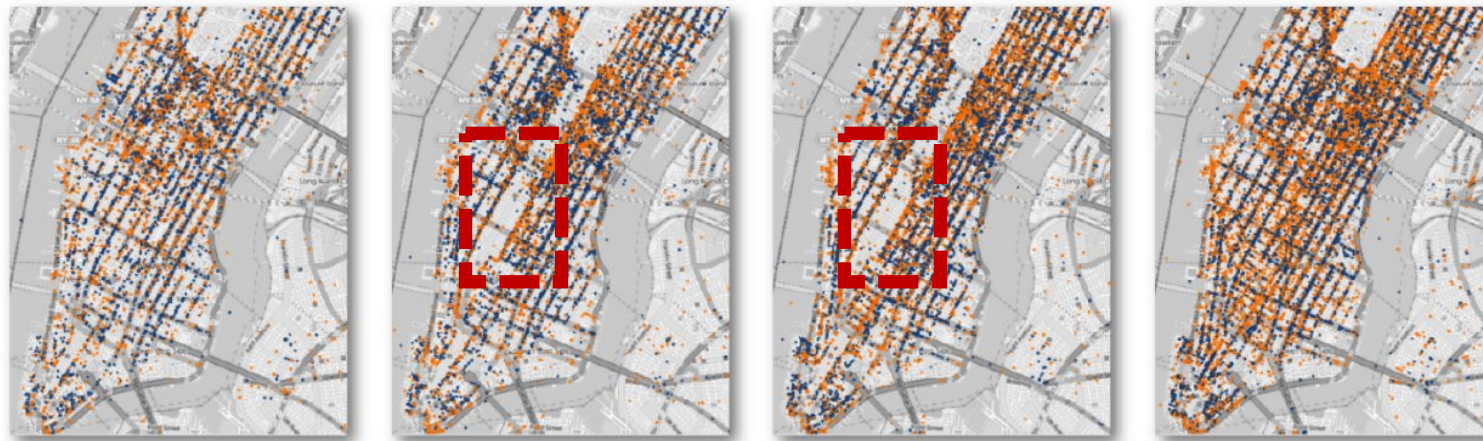
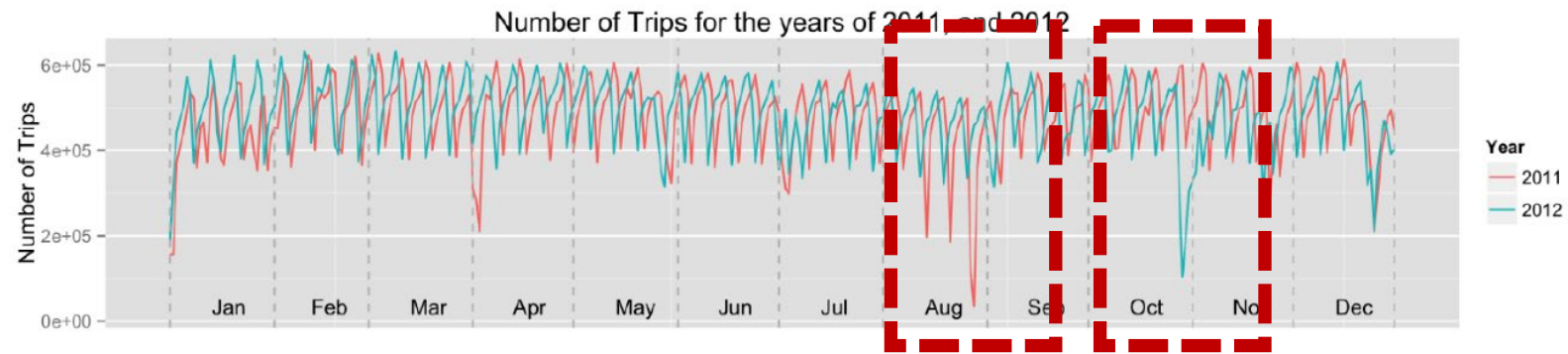
- NYC taxi data:
 - ~175 million trips / year
 - Spatiotemporal
 - Other attributes:
 - Fare, tip
 - Distance
 - Duration
 - ...



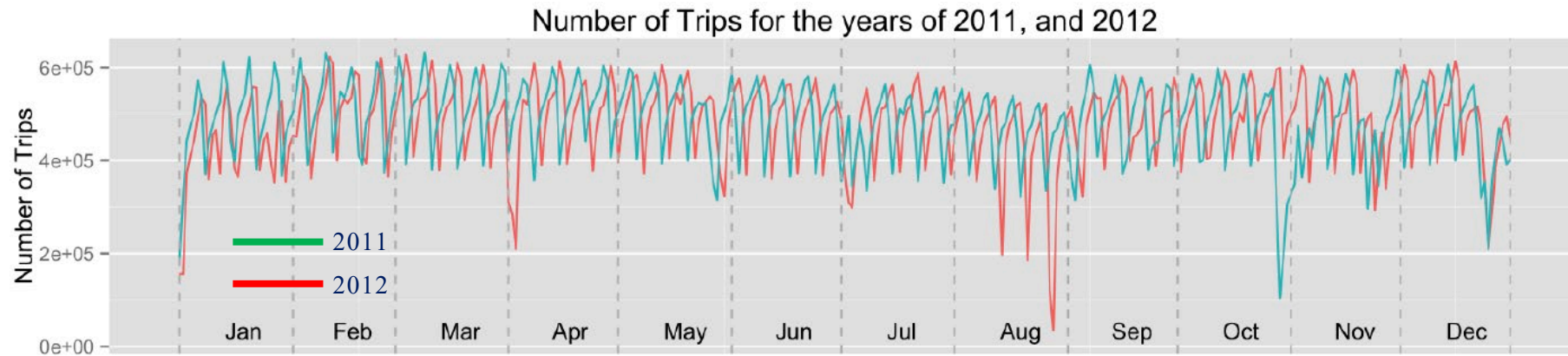
Taxi as sensors of city life

- Understanding the dynamics of the city, how different aspects of the data vary over space and time:
 - *“What is the average trip time from Midtown to the airports during weekdays?”*
 - *“How does the movement changes between Midtown and JFK throughout the day, over different days of the week?”*
 - *“How does the taxi fleet activity vary during weekdays?”*
- Ability to quickly test hypotheses: starting with one query about a specific place (*“movement patterns between Midtown and JFK”*) and generalize to all neighborhoods.

Taxis as sensors of city life



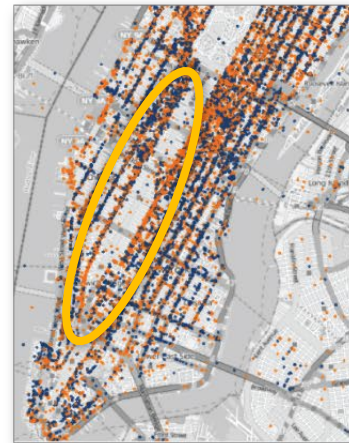
Taxi data slices



7am - 8am



8am - 9am

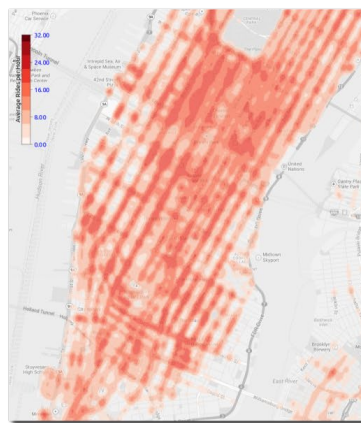
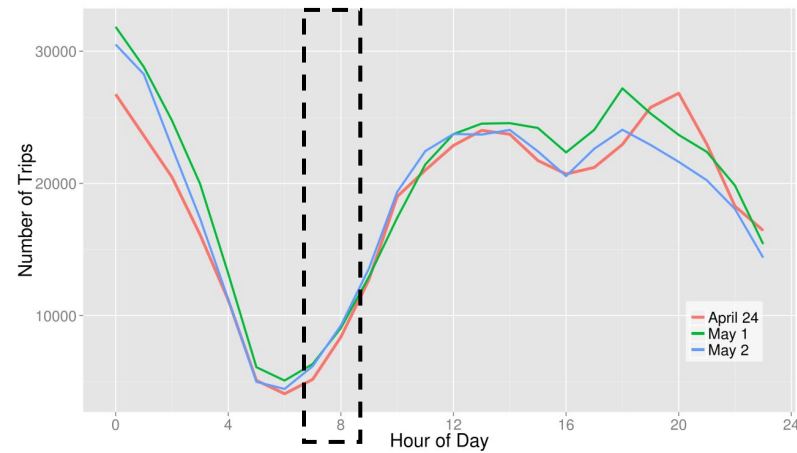


9am - 10am

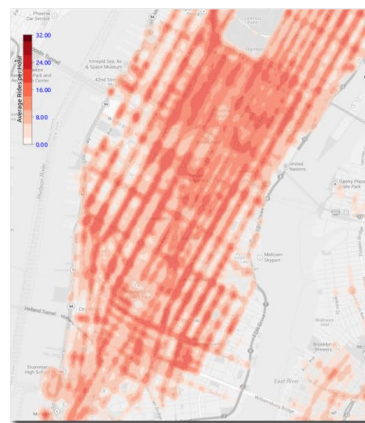


10am - 11am

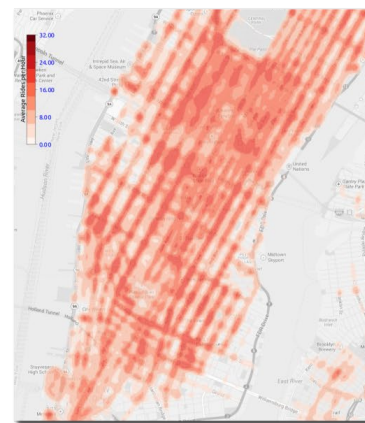
Taxi data slices



April 24



May 1

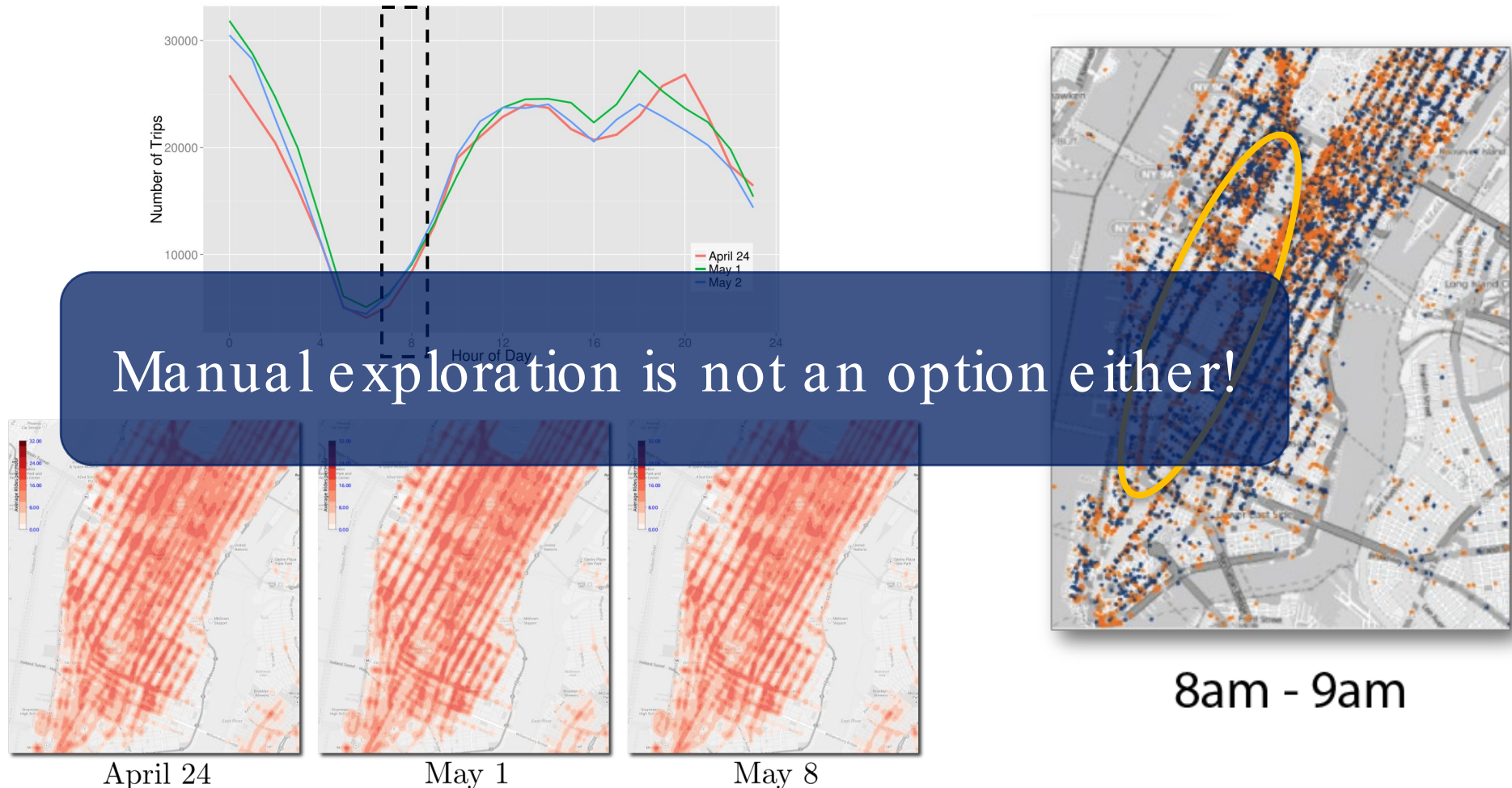


May 8



8am - 9am

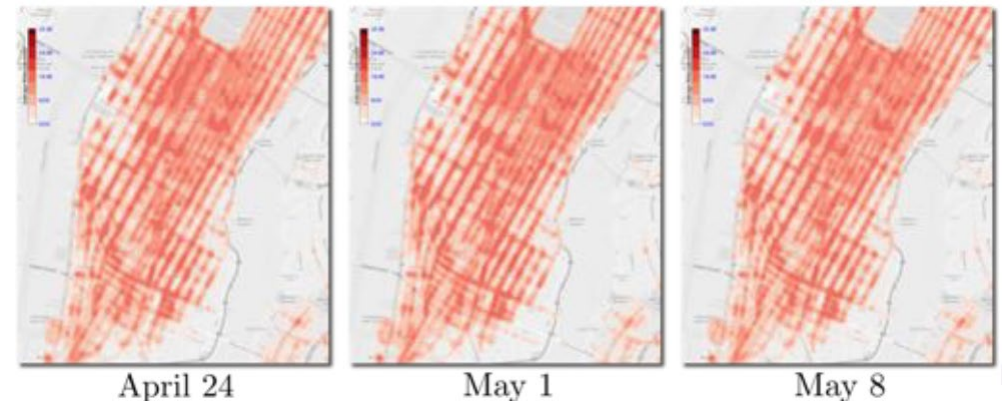
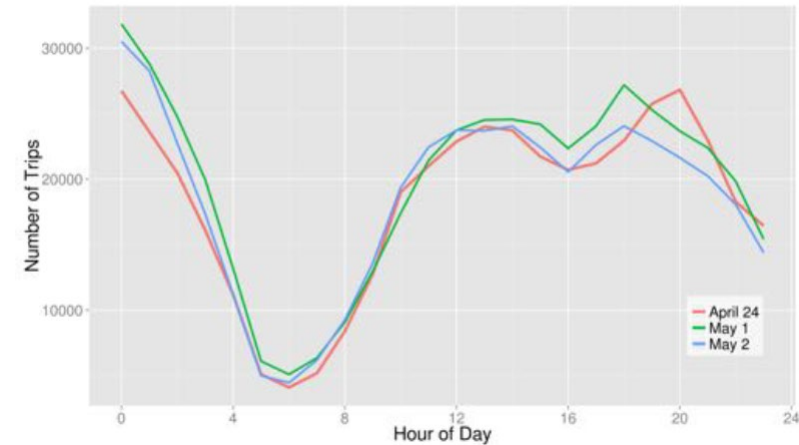
Taxi data slices



Visual analysis process: taxi data



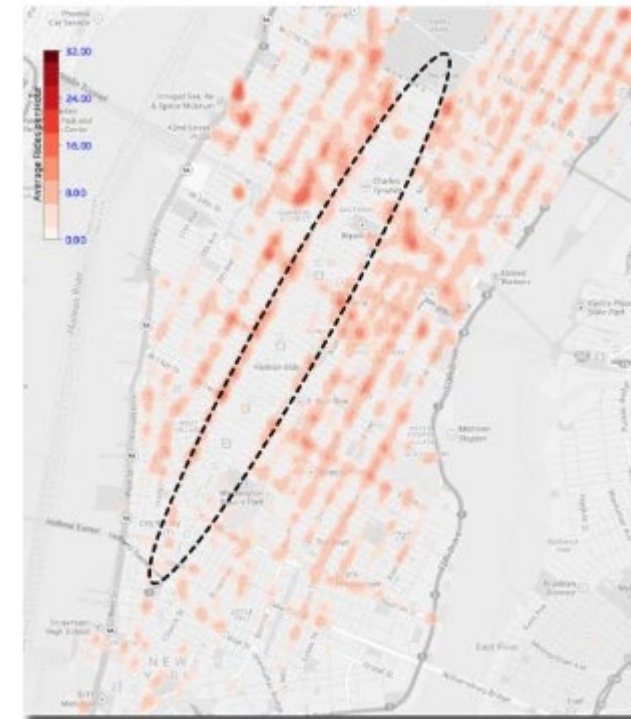
- Too many slices
- 365×24 1-hour slices in just one year
- Which slices are interesting?



Visual analysis process: taxi data



- Too many slices
- 365×24 1-hour slices in just one year
- Which slices are interesting?



May 1 (8-9am)

Usability through visual operations

```
SELECT *  
FROM trips  
WHERE pickup_time in (5/1/11,5/7/11)  
AND dropoff_loc in "Times Square"  
AND pickup_loc in "Gramercy"
```

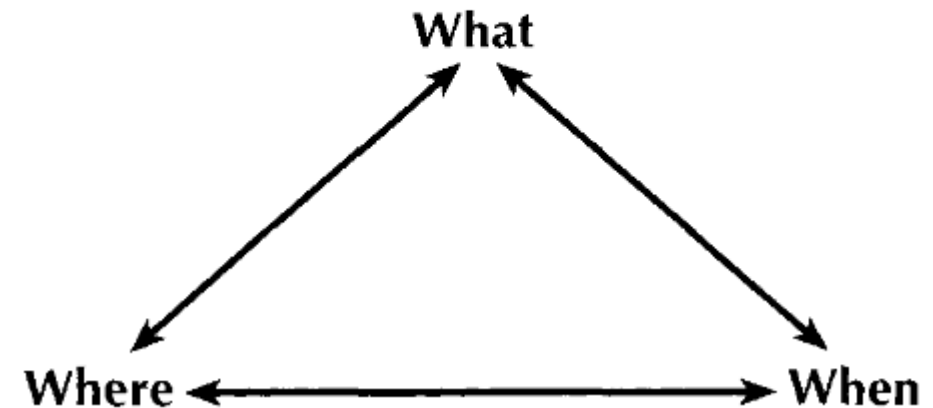
Data selection and result exploration are unified



Visual query model

- Expressive Triad framework:
 - *Where + when* $\rightarrow$ *what* : describe the objects (what) that are present at certain locations and times.
 - *When + what* $\rightarrow$ *where* : describe the locations occupied by objects at a given time.
 - *Where + what* $\rightarrow$ *when* : describe the times that objects occupied a given location.

THE BASIC VIEW COMPONENTS OF THE TRIAD FRAMEWORK



[Peuquet, 1994]

Visual query model

- Expressive Triad framework:
 - Where + when* $\rightarrow$ *what* : “What is the average trip time from Midtown to the airports during weekdays?”
 - When + what* $\rightarrow$ *where* : “Where are the hot spots in Manhattan in the weekends?”
 - Where + what* $\rightarrow$ *when* : “When were activities restored in Lower Manhattan after the Sandy hurricane?”

Where? When?
What?



Comparison of taxi trips: characterizing neighborhoods

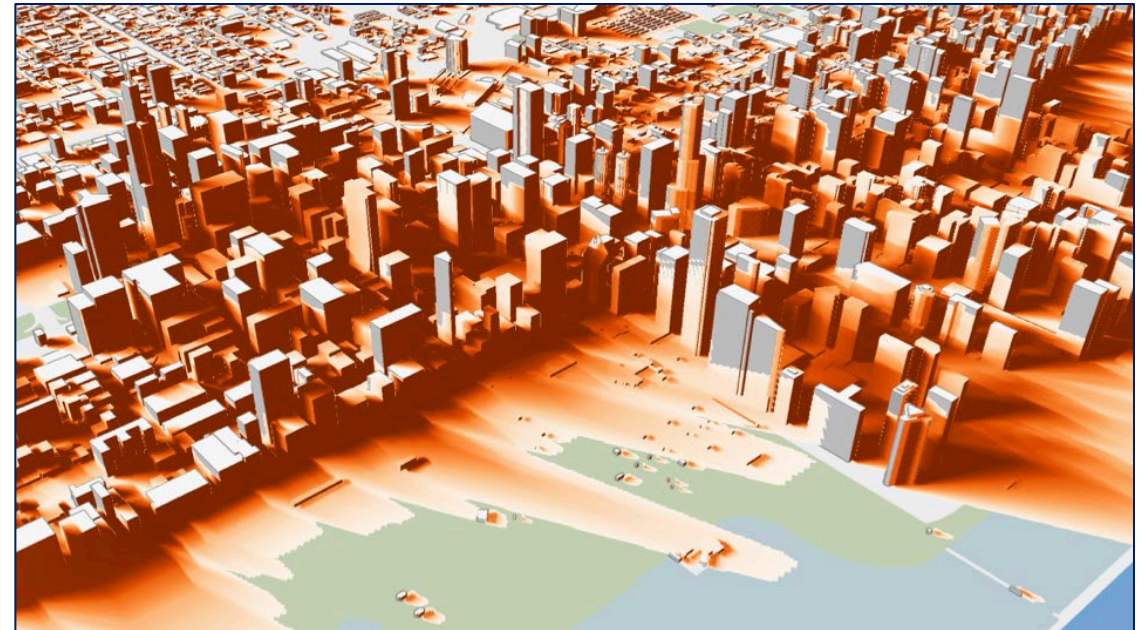


- Midtown and Upper East are the most active areas.
- Midtown is a popular destination in the morning and downtown in the night.
- Over the weekend, increase number of drop-offs in Downtown.
- Harlem is underserved by taxis.

Ability to quickly test hypotheses: starting with one query about a specific place and generalize to all neighborhoods.

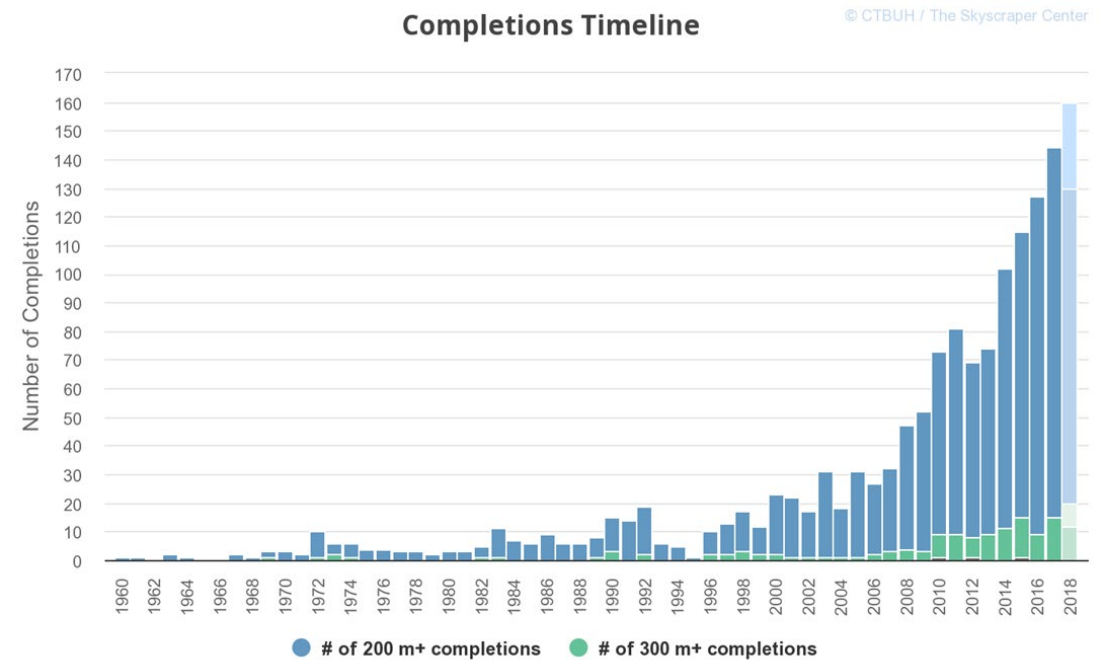
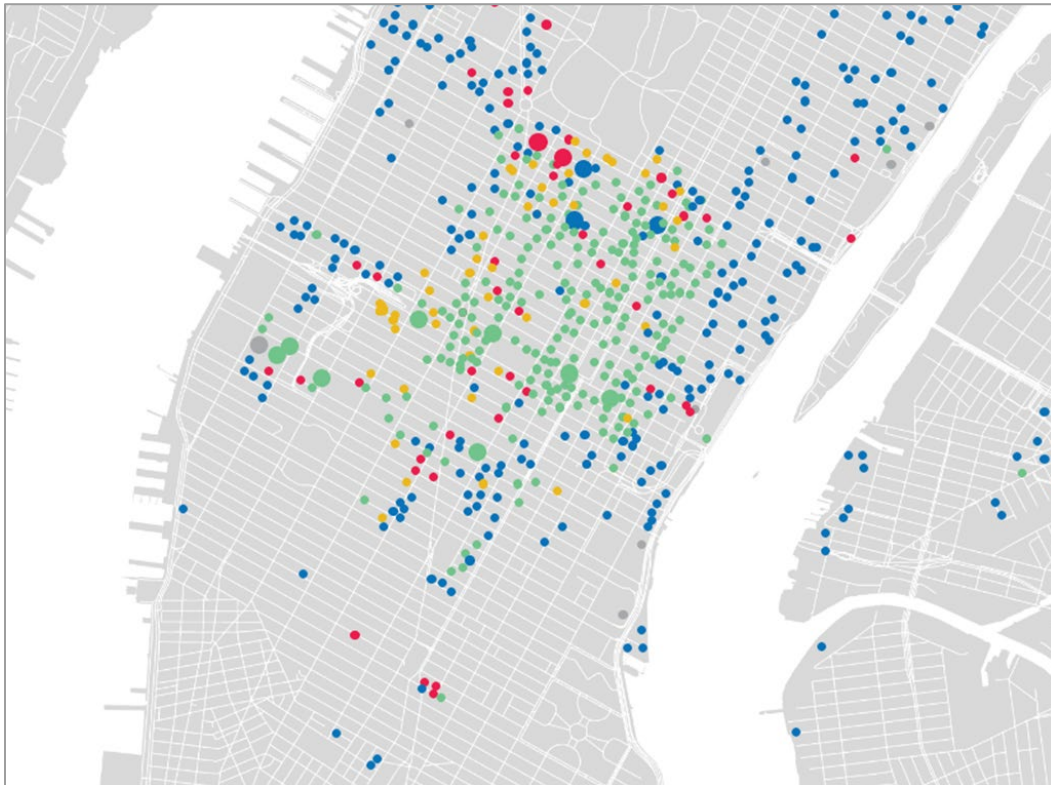
Shading data at street level

- Why shadow / shading information?
 - Identify the potential places with higher risk of **black ice formation**.
 - Improve the pedestrian level of comfort by identifying places with more **shade in summer**.



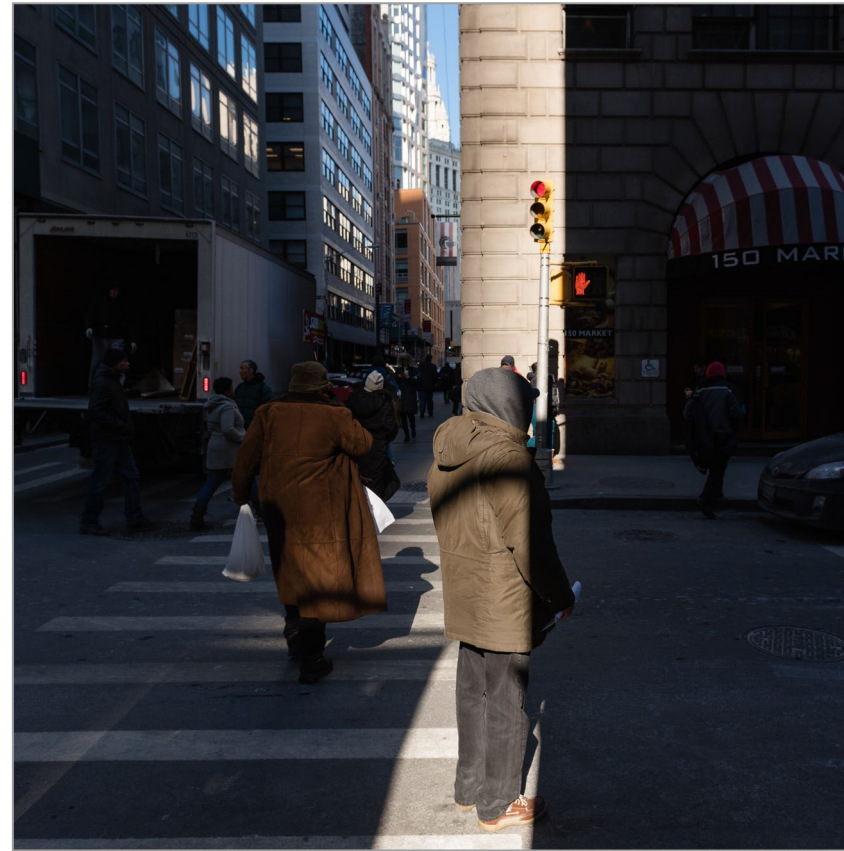
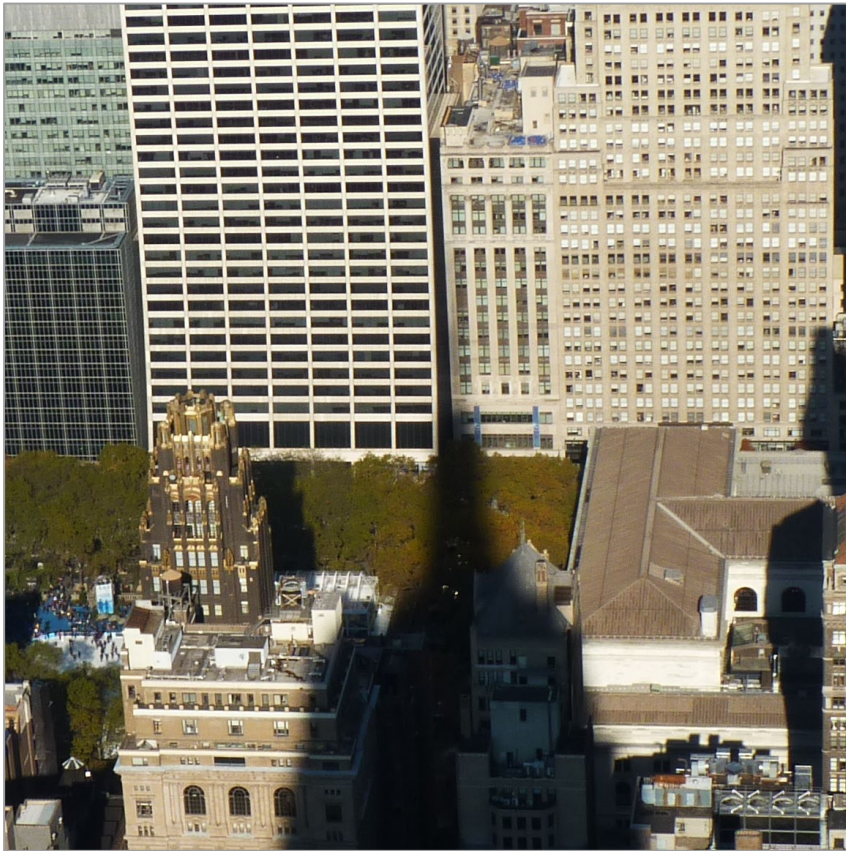
Shadow Accrual Maps: Efficient Accumulation of City-Scale Shadows over Time, TVCG 2019

Skyscrapers



Source: Skyscraper Center

Shadow impact



Shadow dispute



Meet: Columbus Circle, Sunday, November 8, 2015, 10:30am.
March begins at 11:00 am sharp—east on 57th St. to Sutton Place. (Rain or shine!)

Bring a black umbrella to symbolize the shadows the towers will cast.

Why:
Central Park, New York City's front yard, is under siege by developers building super-tall towers that will cast 1,000-foot shadows over the park...castles in the sky for the privileged few at the expense of everyone else's sunshine.

In a city where small businesses can be choked by red tape, developers can legally build higher than the Freedom Tower without asking anyone's permission. Because zoning laws were written before architects could build sliver towers this high, developers can simply buy air rights and use empty floors to work around height limits.

It's time to update our zoning laws!

We are not anti-development. We are pro smart-development, taking into account community needs and the protection of green spaces, public parks and waterways across New York City.

We're calling on our elected officials to issue a temporary moratorium on the construction of towers over 600 feet until city and community leaders can work with developers to enact reasonable changes to zoning laws.

Facebook.com/StandAgainstTheShadows @NYC_Shadows
#StandAgainstTheShadows

Boston developer wants exception to shadow law

Tall order is sought for tower proposal

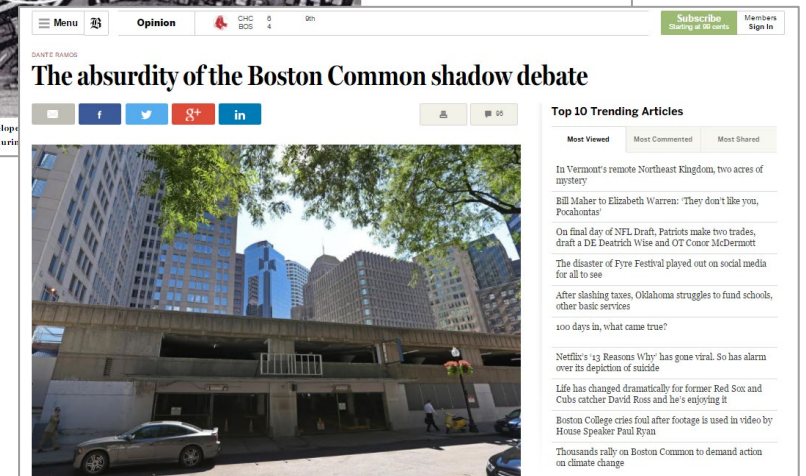
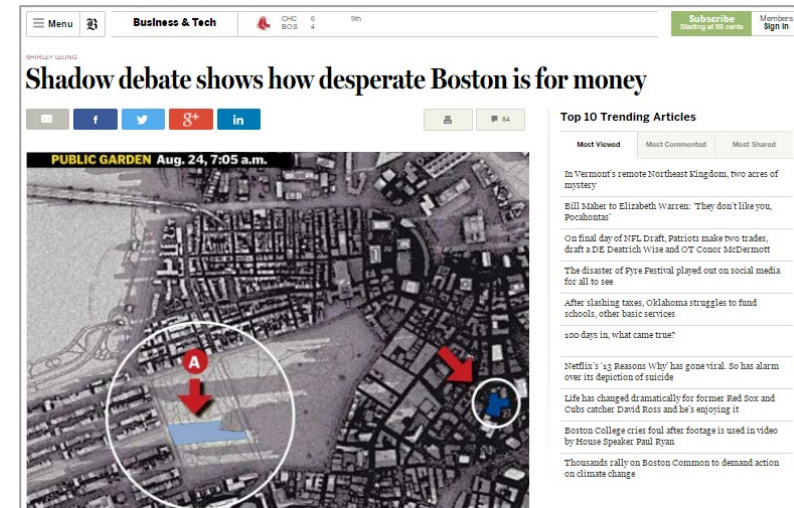
Donna Goodison Monday, April 10, 2017



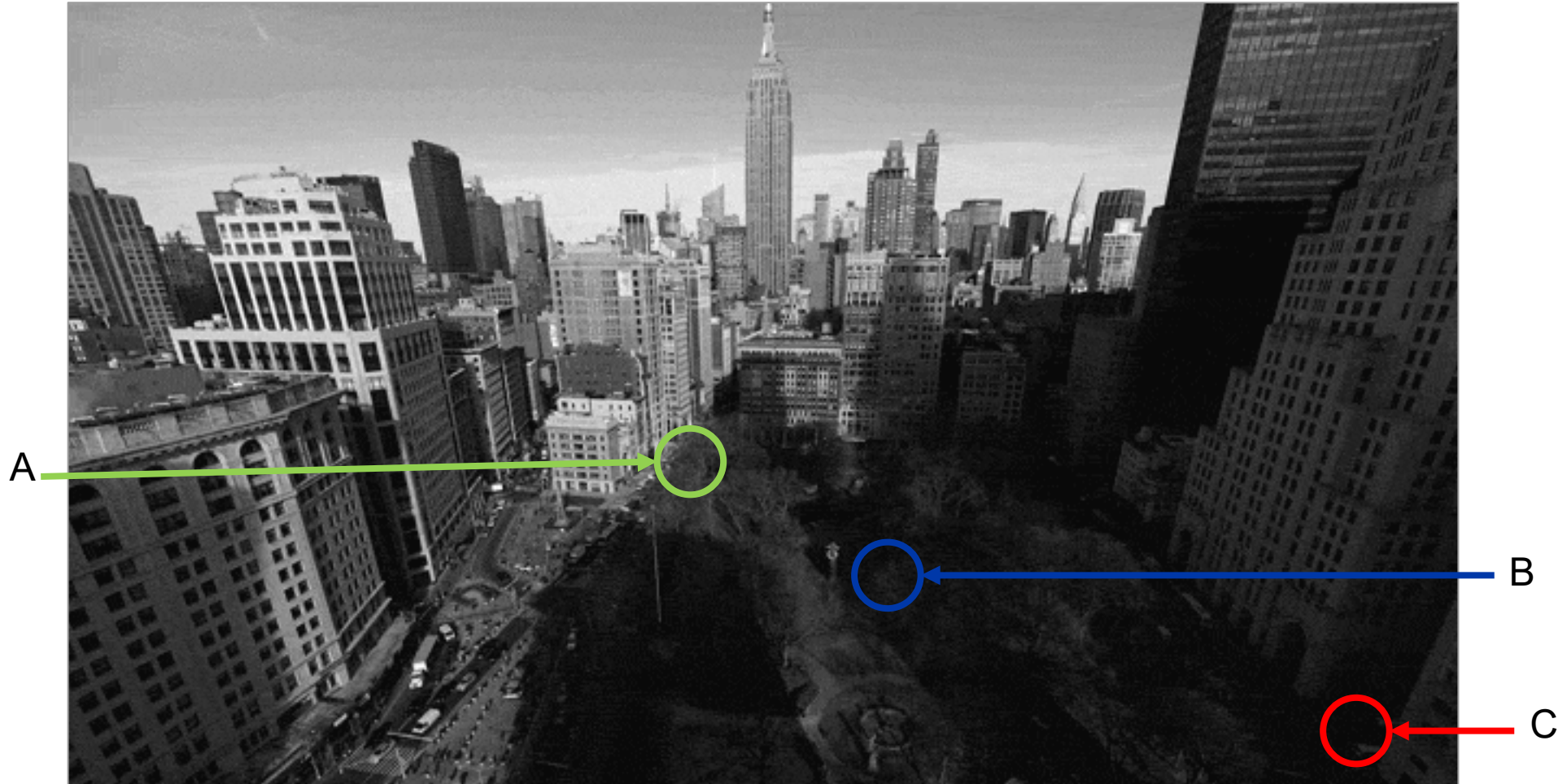
Boston City Council votes in favor of changing 'shadow' law

Council backs shade

Dan Atkinson Thursday, April 27, 2017



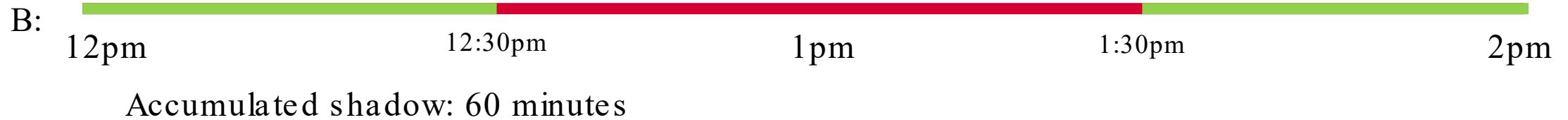
Quantifying shadow



~2 hours time lapse

Mapping the Shadows of New York City- The New York Times

Quantifying shadow



Objectives

Support the comprehensive **interactive analysis** of **city-scale shadows**;
Accumulated shadow.

Enable analyses considering **large periods** (e.g. months);
Amount of shadow in Central Park during summer vs. amount of shadow in Central Park during winter.

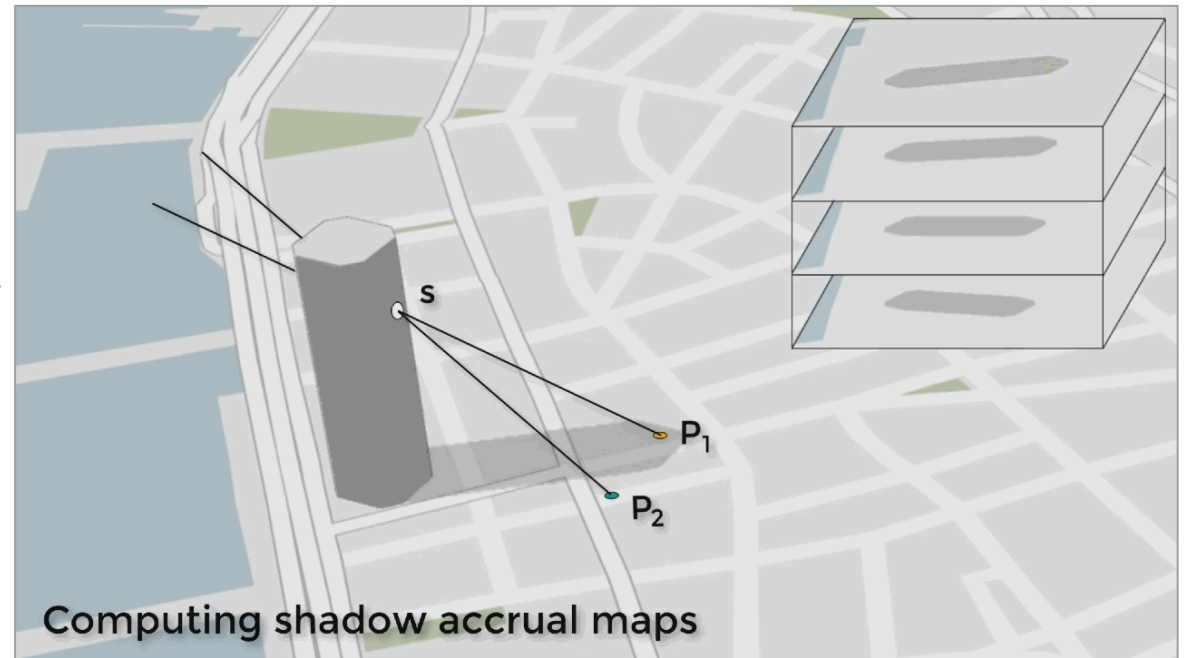
Different **analysis measures**;
Differentiate between shade and shadow:
-Shade: positive during hot summers.
-Shadow: negative during cold winters.
Shadow contribution of each building.

Shadow Accrual Map

3D texture where each 2D slice is a timestep.

Consider shadows cast by s at timestep t_1 (e.g. 12 pm) and t_2 (e.g. 1pm) being p_1 and p_2 .

The values between t_1 and t_2 will be linearly interpolated between p_1 and p_2 and stored in the corresponding 2D slice.



Shadow Accrual Map is the same as computing shadow maps for every step (proof in the paper)

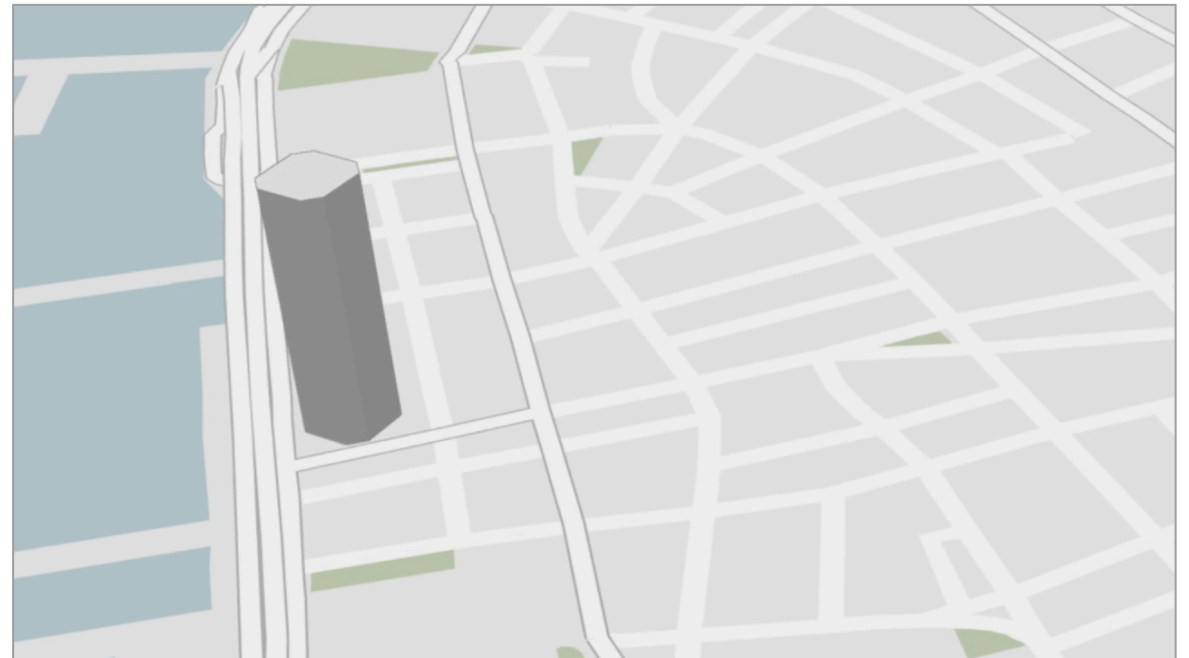
Inverse Accrual Maps

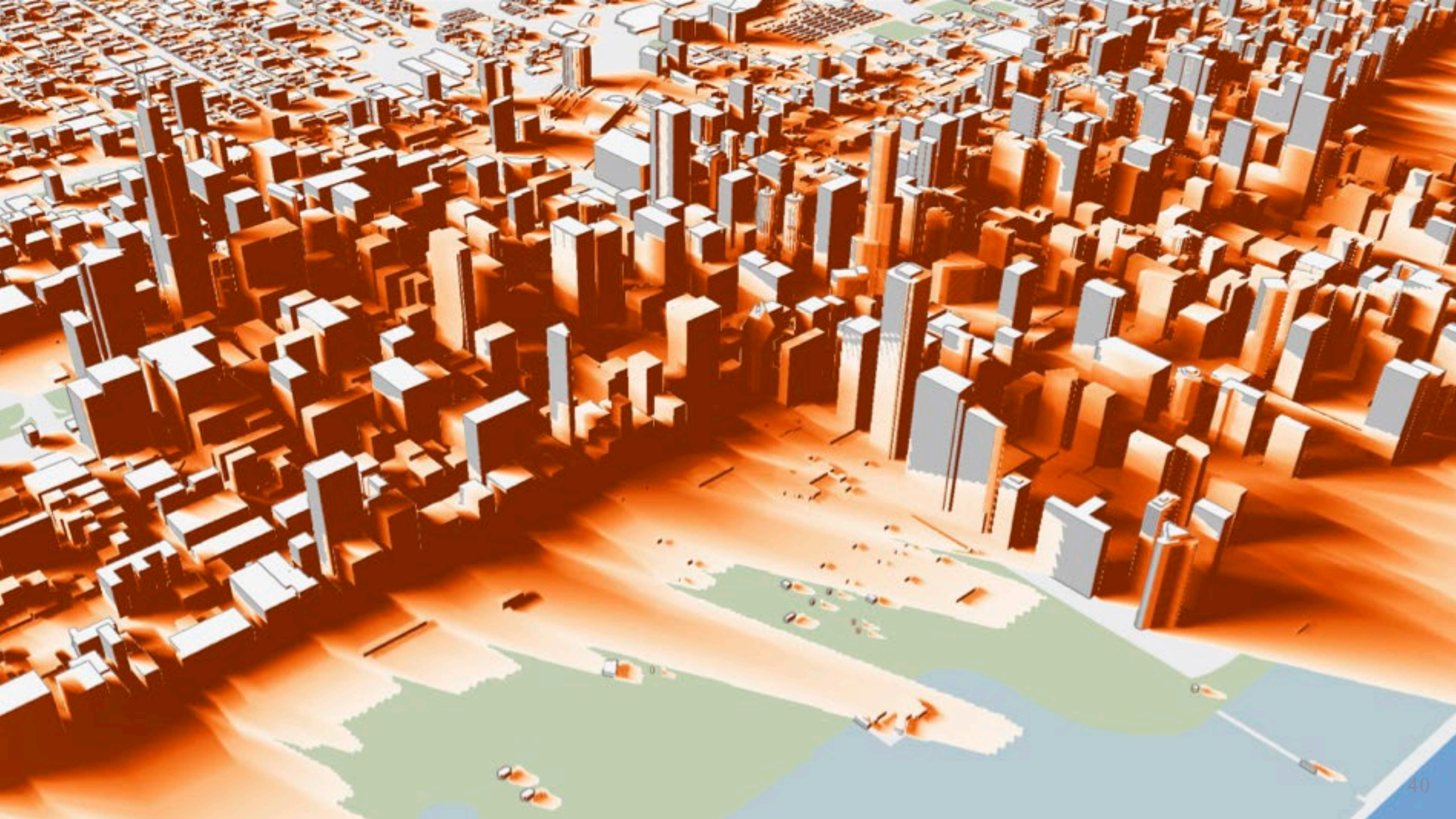
Ray tracing-based.

More accurate than Shadow Accrual Maps.

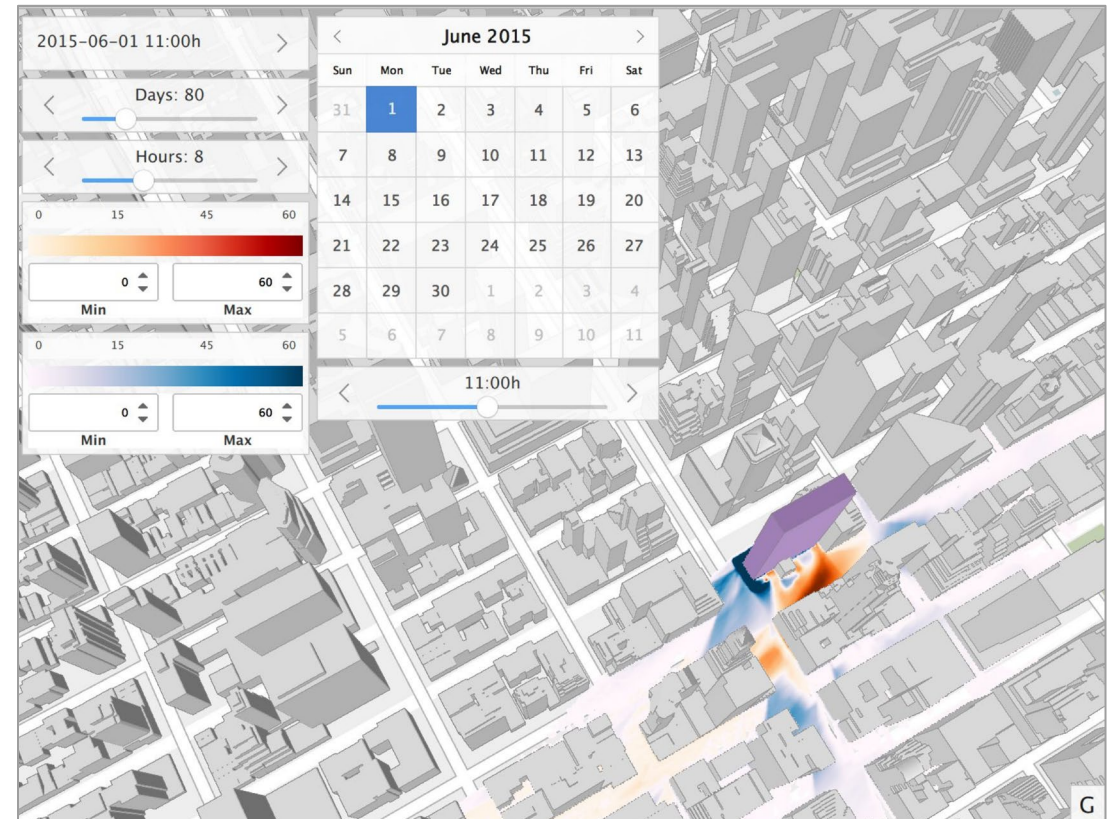
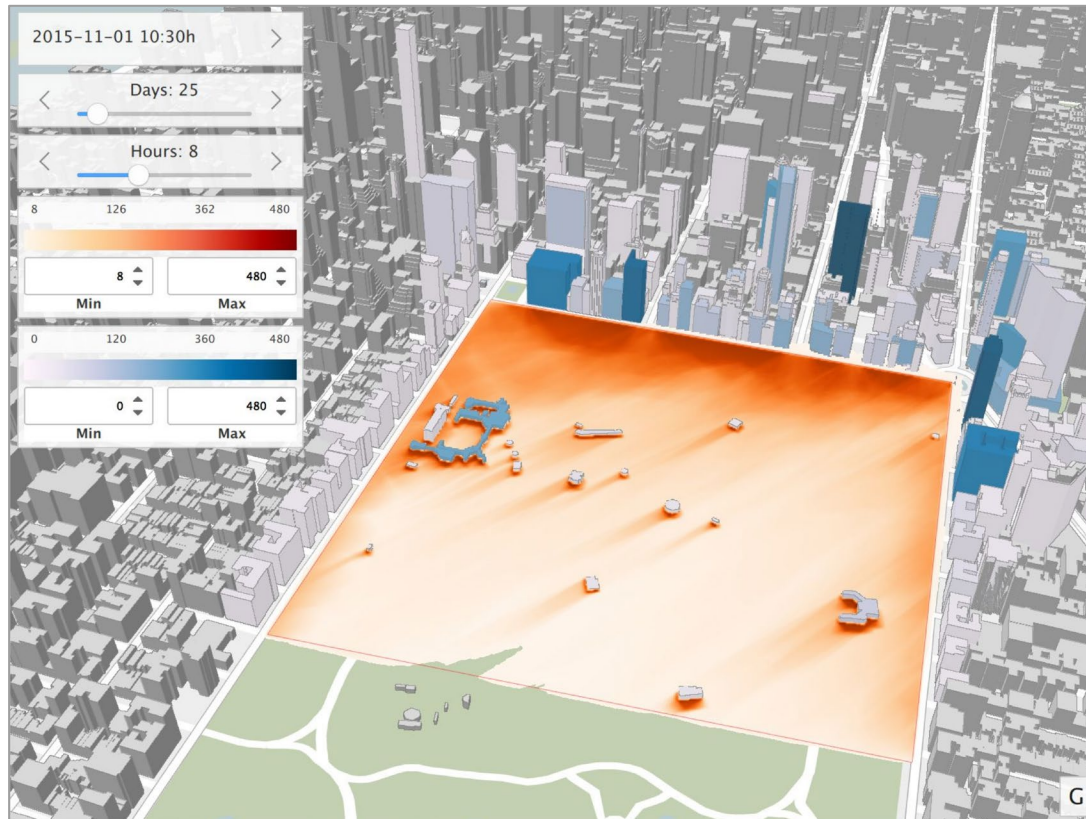
Used for detailed shadow analysis.

Same principle of considering linear sun movement.



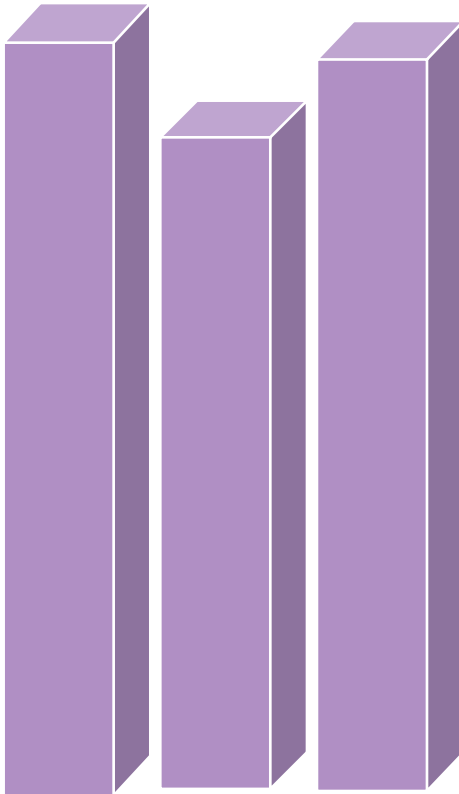


Shadow Profiler

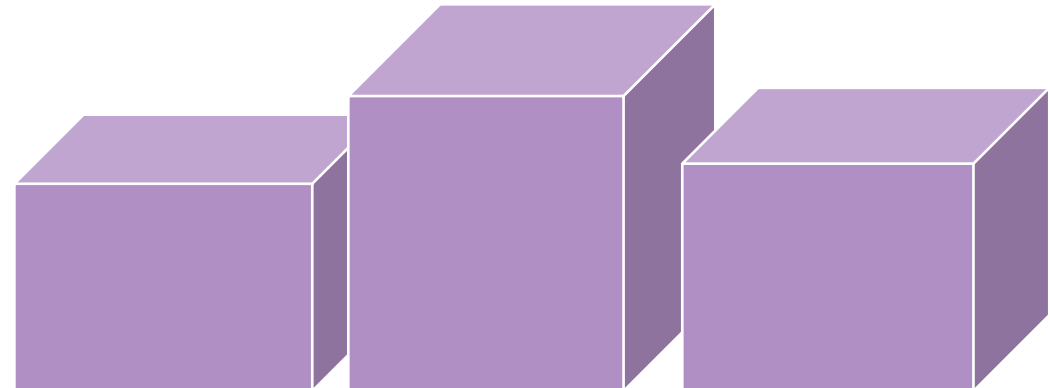


Shadow impact of two scenarios

Analysis of the impact of two different scenarios, with buildings with **same total floor area**



Scenario 1: Lean and tall buildings



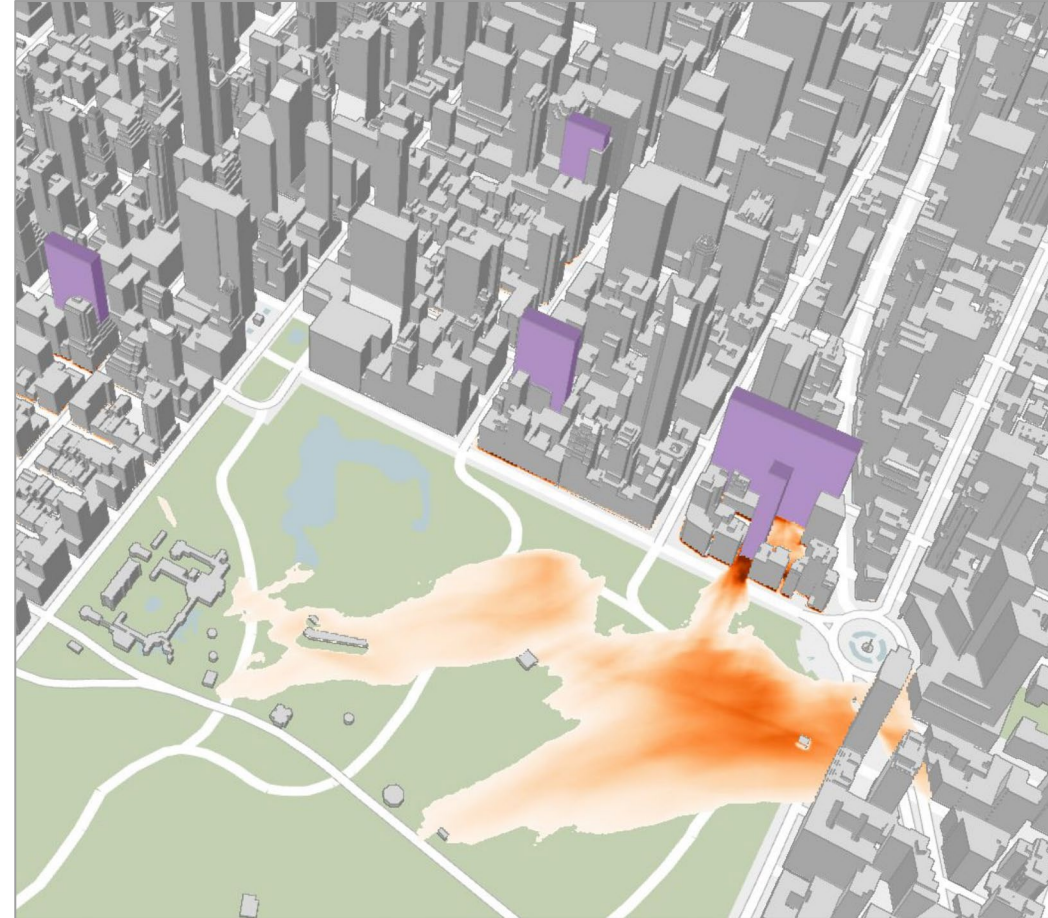
Scenario 2: Bulky and short buildings



Shadow impact of two scenarios



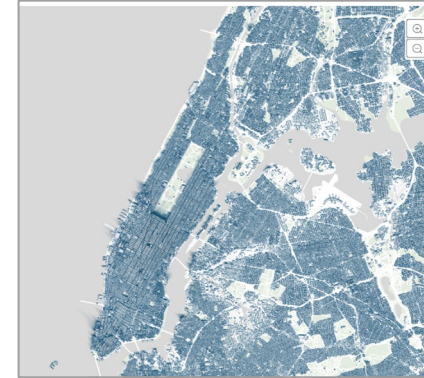
Shadow impact of scenario 1



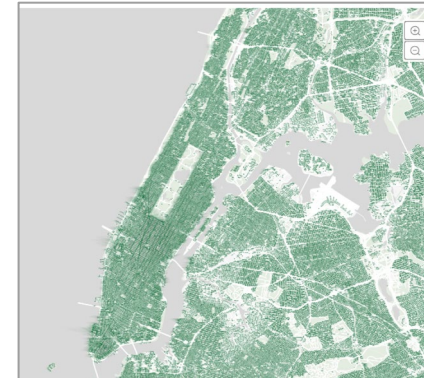
Shadow impact of scenario 2



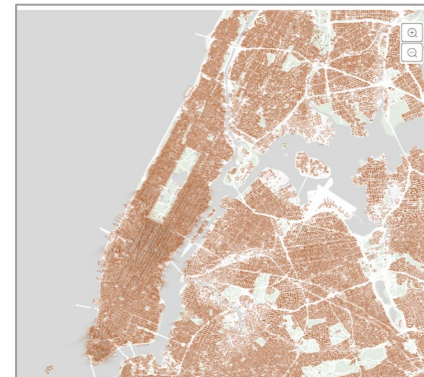
Winter



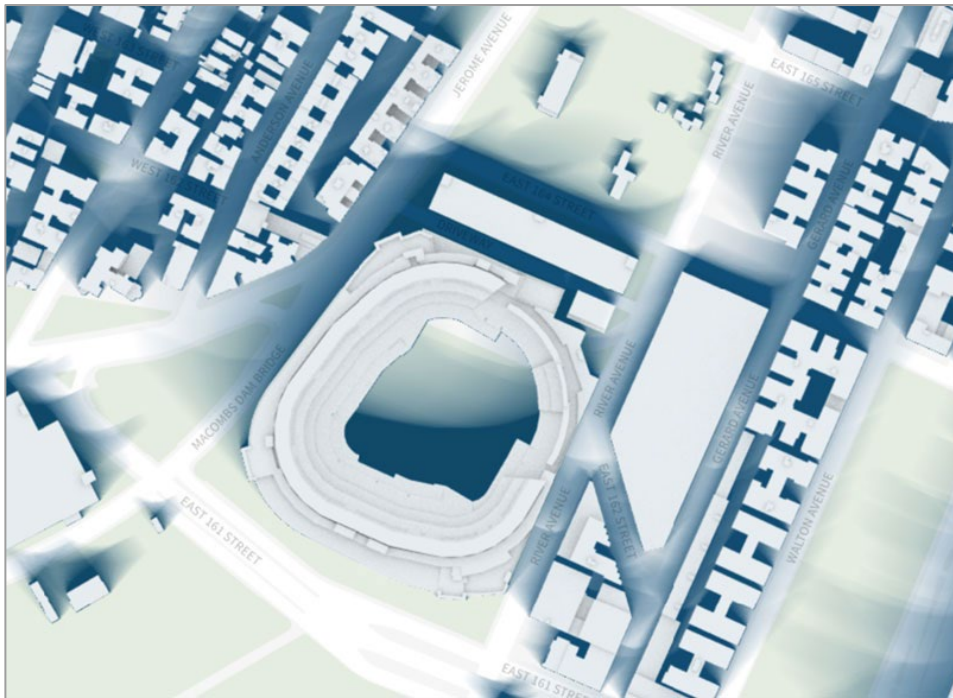
Spring /
Fall



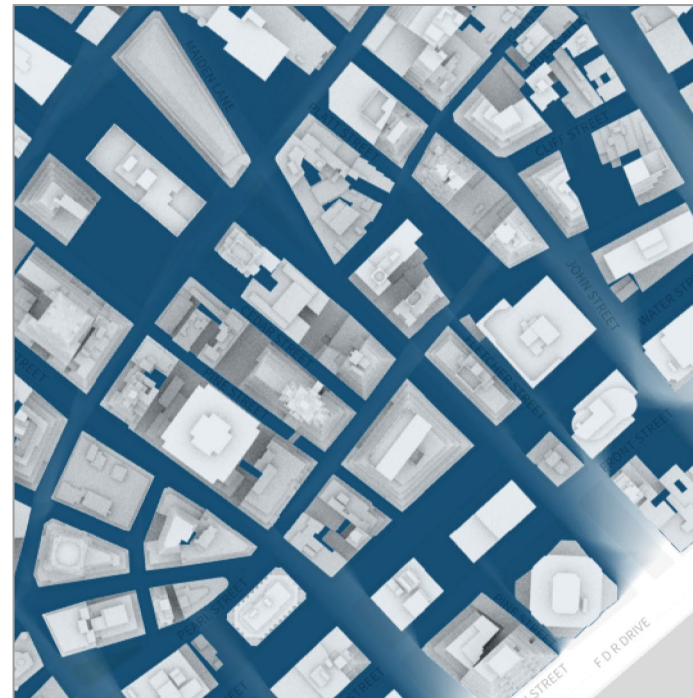
Summer



Mapping the shadows of New York City



Yankee Stadium



Financial District



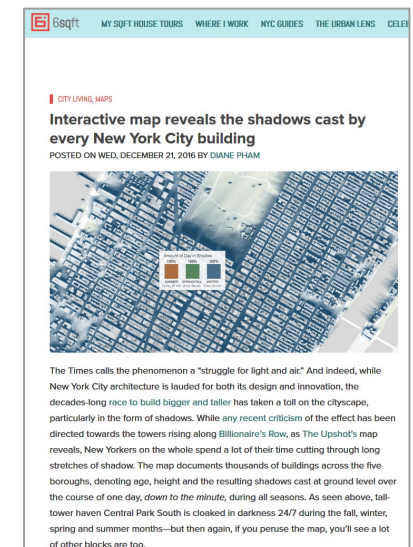
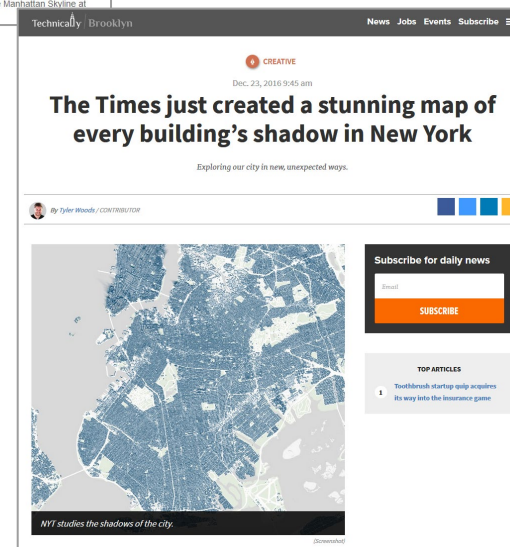
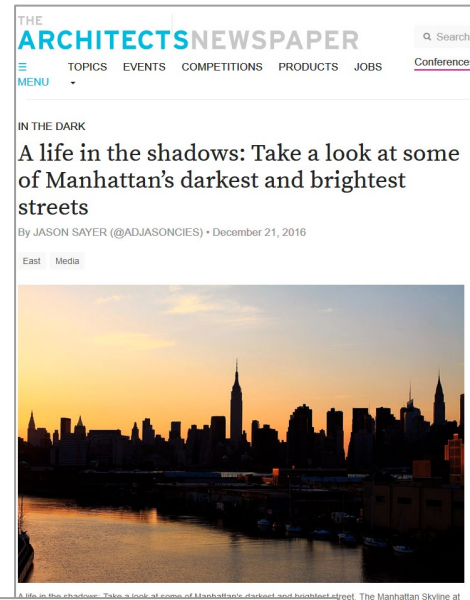
Broadway Avenue

Mapping the shadows of New York City



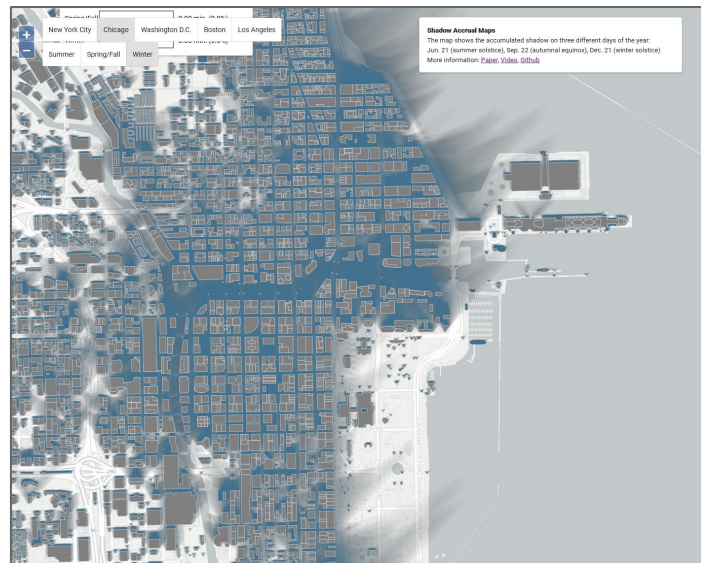
New York Times

<https://nyti.ms/2k0bF0G>

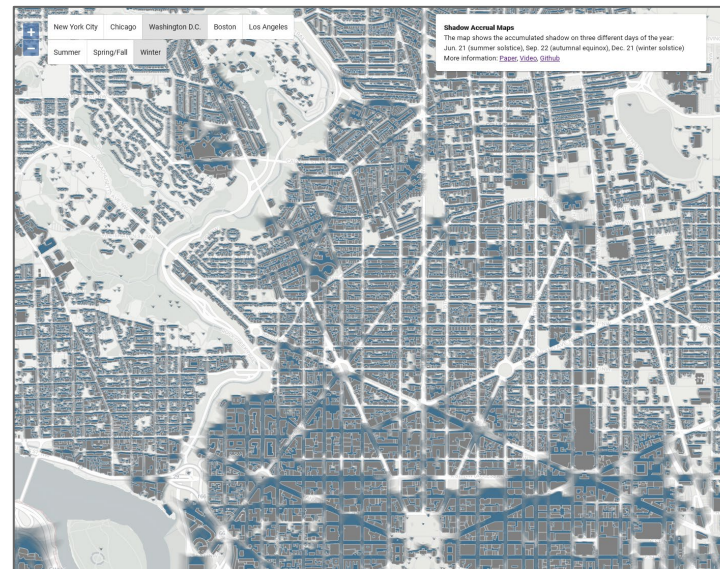


Mapping the shadows across the US

Chicago



Washington DC



Los Angeles



Boston



<https://github.com/VIDA-NYU/shadow-accrual-maps/>

Urban Mosaic: street-level images

“... a tool for exploring the urban fabric through a spatially and temporally dense data set of 7.7 million street-level images from New York City, captured over the period of a year. Working in collaboration with professional practitioners, we use Urban Mosaic to investigate questions of accessibility and mobility, and preservation and retrofitting.”

Urban Mosaic: Visual Exploration of Streetscapes Using Large-Scale Image Data

Fabio Miranda¹, Maryam Hosseini², Marcos Lage³, Harish Doraiswamy¹,
Graham Dove¹, Cláudio T. Silva¹

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ABSTRACT

Urban planning is increasingly data driven, yet the challenge of designing with data at a city scale and remaining sensitive to the impact at a human scale is as important today as it was for Jane Jacobs. We address this challenge with Urban Mosaic, a tool for exploring the urban fabric through a spatially and temporally dense data set of 7.7 million street-level images from New York City, captured over the period of a year. Working in collaboration with professional practitioners, we use Urban Mosaic to investigate questions of accessibility and mobility, and preservation and retrofitting. In doing so, we demonstrate how tools such as this might provide a bridge between the city and the street, by supporting activities such as visual comparison of geographically distant neighborhoods, and temporal analysis of unfolding urban development.

Author Keywords

Urban planning; Interactive visualization; Data analysis; Urban data

CCS Concepts

•Human-centered computing → Human computer interaction (HCI); Visual analytics; Visualization toolkits;

“A sense of place is built up, in the end, from many little things too, some so small people take them for granted, and yet the lack of them takes the flavor out of the city.”

(Jane Jacobs, Downtown is for People)

INTRODUCTION

For those of us living in or visiting the world’s major cities, their dynamism and complexity are immediately apparent. Yet urban planners and designers must work in a context where any single intervention, perhaps aimed at altering just one aspect, can have a wide ranging impact on a variety of interrelated components [12], affecting things at both a macro *city scale* and a micro *human scale*. Examples of changes at a city scale

include, suburbanization, economic deconcentration, modification to transport infrastructure, rezoning and/or gentrification of neighborhoods, and major renewal projects. Examples of changes at a human scale, on the other hand, are reflected in the city’s urban fabric, and include aspects that make a city livable, encourage walking, and contribute to the perception of safety; in other words affect the day to day lives of its inhabitants. This might be manifested in lighting, shadow, sky exposure, open-front shops, the details of building facades, etc. [32].

Responding to these challenges at the city scale, planners and designers have been aided by a rapid growth in data from urban environments, and so are able to turn to computational methods and large-scale data analysis, which increase understanding by quantifying different aspects of the city (e.g., [36, 13, 14, 29, 70, 67, 48]). However, while sensitivity to the impact of change at a human scale remains as important today as it was for Jane Jacobs and others in the 1950s and 1960s [47], analyses of suitable data, which emphasize qualitative, visual details, are often difficult and time-consuming to perform. Although there is an increasing availability of street-view images, which support a degree of virtual assessment and auditing of the built environment, their distribution is often temporally sparse and so analysis is limited.

This paper introduces Urban Mosaic, a tool for visually exploring the urban fabric. It responds to the challenges practitioners face by employing a newly available spatially and temporally dense data set of street-level images from New York City (NYC) Urban Mosaic is a visual exploration system designed to help practitioners in urban planning and design gain insight into the human scale impact of changes in the urban fabric. It utilizes state-of-the-art computer vision techniques for image similarity search and clustering, together with efficient spatio-temporal selection and aggregation over the image metadata to visually explore and map this image data set. It further allows the analysis to be augmented using spatio-temporal urban data from a variety of other sources (e.g., census, transport, crime, weather, housing market, zoning, noise complaints). The image data set used in this work contains 7.7 million images captured in the Manhattan and Brooklyn boroughs between April 2016 and April 2017 using car-mounted cameras. Urban Mosaic has been developed as a collaboration between researchers in urban planning, visual analytics, and HCI. We include a detailed

[Miranda et al., 2020]

Urban Mosaic: street-level images

- Demonstrates the potential for images from large-scale street-view data sets to help bridge data-driven urban planning and design at city scale and at the human level.
- Exploration of the urban fabric using large spatiotemporal collection of images.
- Visual comparison of geographically distant areas to analyze unfolding urban developments.

Urban Mosaic: Visual Exploration of Streetscapes Using Large-Scale Image Data

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ABSTRACT

Urban planning is increasingly data driven, yet the challenge of designing with data at a city scale and remaining sensitive to the impact at a human scale is as important today as it was for Jane Jacobs. We address this challenge with Urban Mosaic, a tool for exploring the urban fabric through a spatially and temporally dense data set of 7.7 million street-level images from New York City, captured over the period of a year. Working in collaboration with professional practitioners, we use Urban Mosaic to investigate questions of accessibility and mobility, and preservation and retrofitting. In doing so, we demonstrate how tools such as this might provide a bridge between the city and the street, by supporting activities such as visual comparison of geographically distant neighborhoods, and temporal analysis of unfolding urban development.

Author Keywords

Urban planning; Interactive visualization; Data analysis; Urban data

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[Miranda et al., 2020]

Temporally -dense street -level images



Objectives

Support the **interactive** analysis of the city at the micro scale, over geographically distant regions.



Comparison of the urban fabric in different regions



Assessment of features in the built environment



Assessment of walkability and accessibility

Image query composition

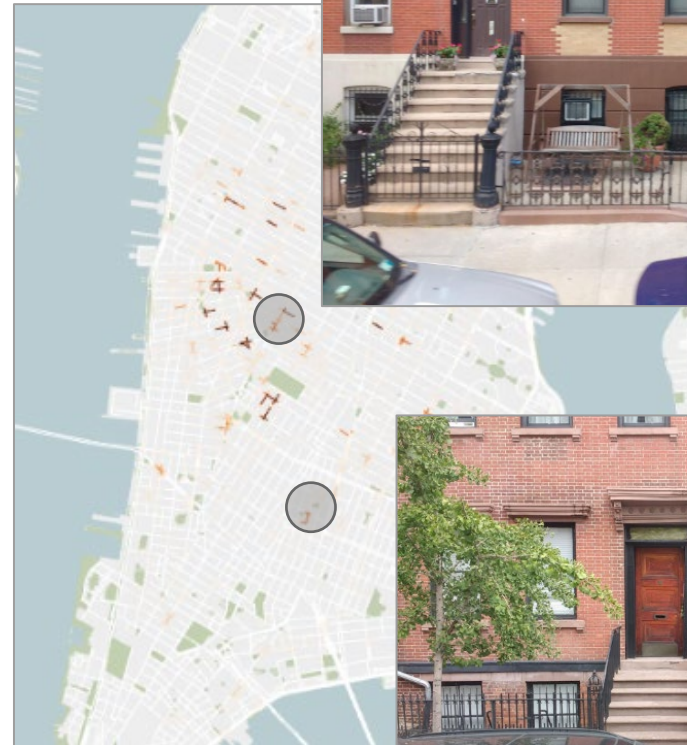
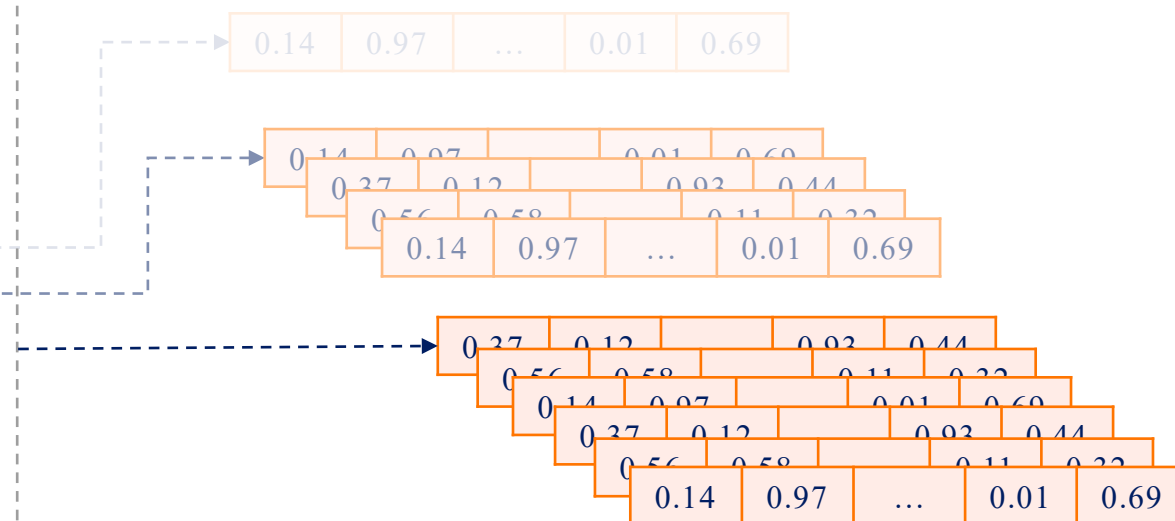


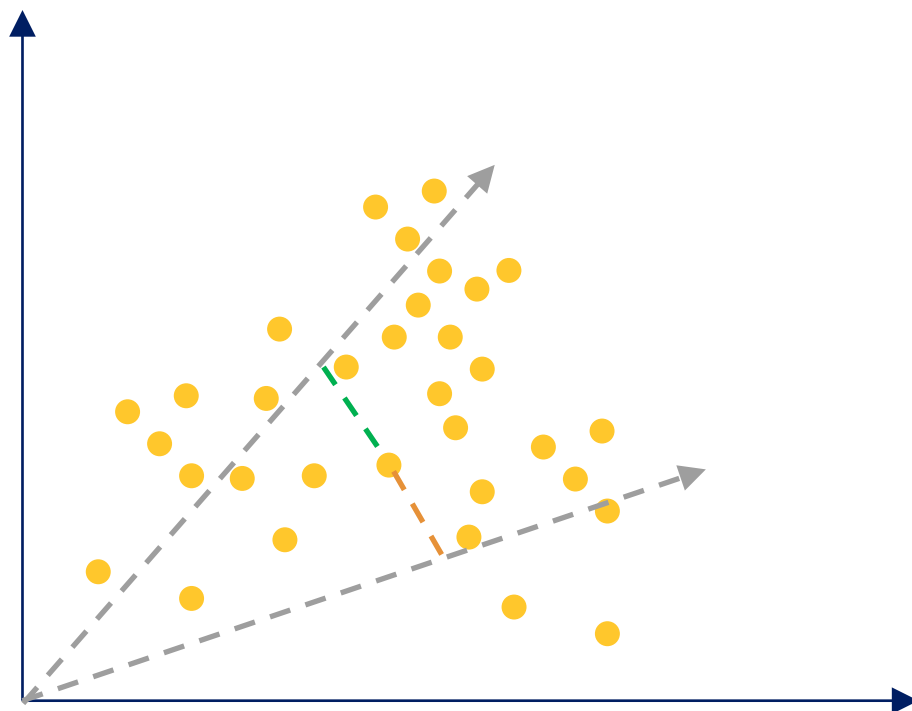
Image embeddings



21 embeddings per image

2.5 TB worth of embeddings data

Locality sensitive hashing



0.37	0.12	...	0.93	0.44
------	------	-----	------	------

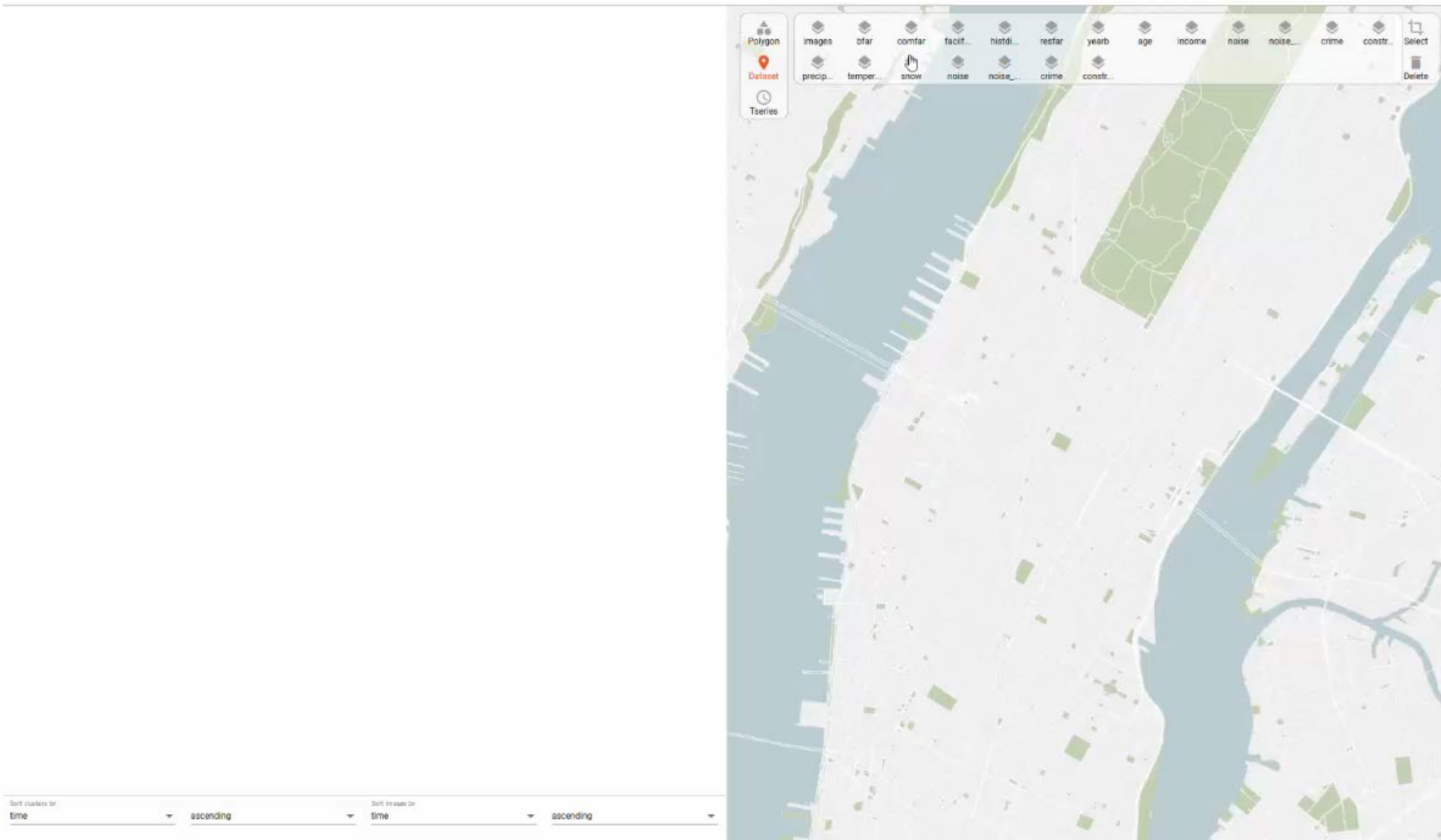
4 kbytes per embedding

0	0	...	1	0
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64 bytes per embedding

$$\alpha_{1,2} = \cos^{-1}\left(\frac{\vec{v}_1 \cdot \vec{v}_2}{|\vec{v}_1||\vec{v}_2|}\right)$$

21 GB worth of embeddings data



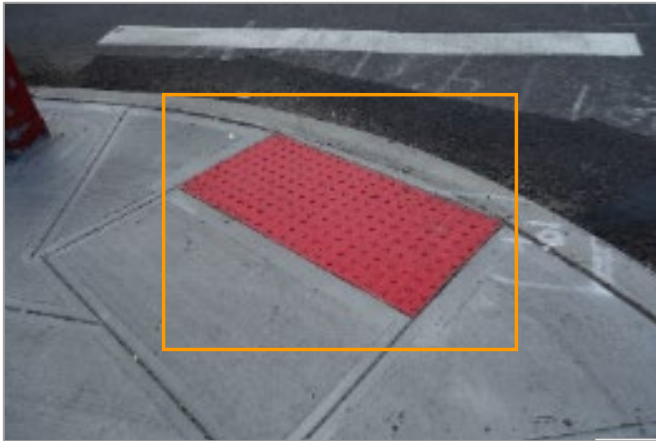
Accessibility for older adults



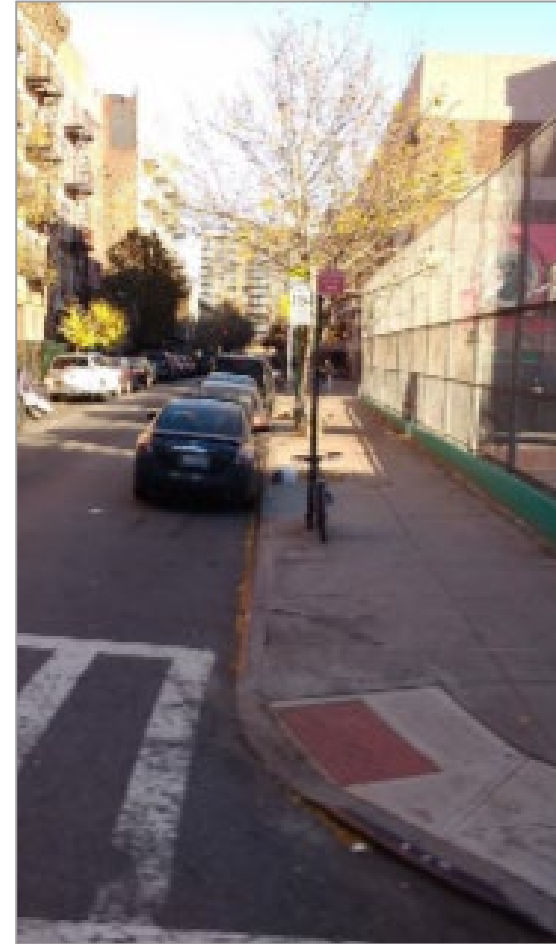
Focuses on interventions that enable older adults to “Age in Place” and prevent outdoor fall.

Impact of the neighborhood environment on falls.

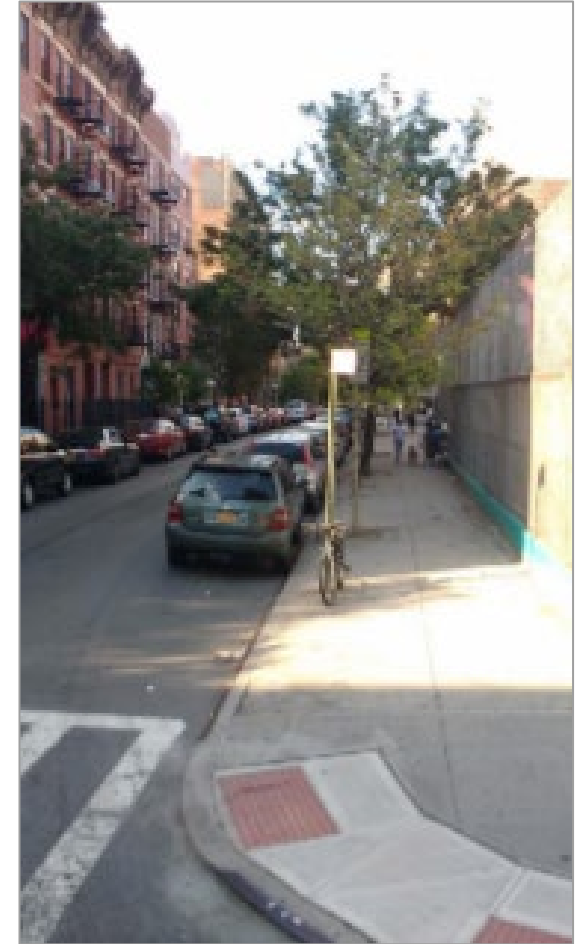
Accessibility: installation of tactile pavings



June

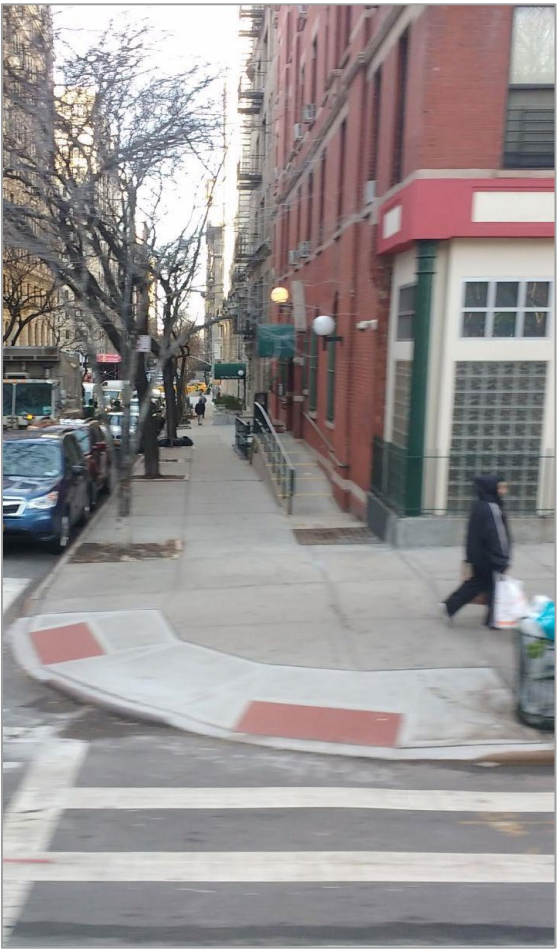
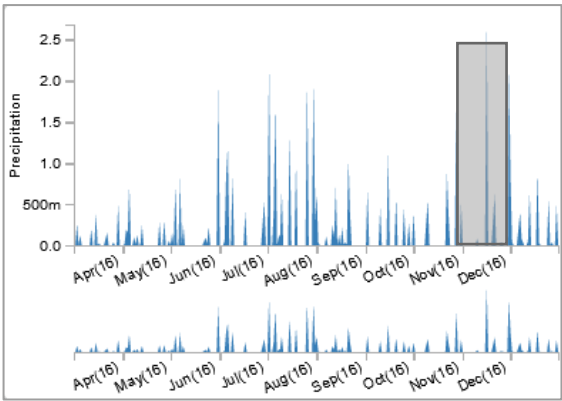
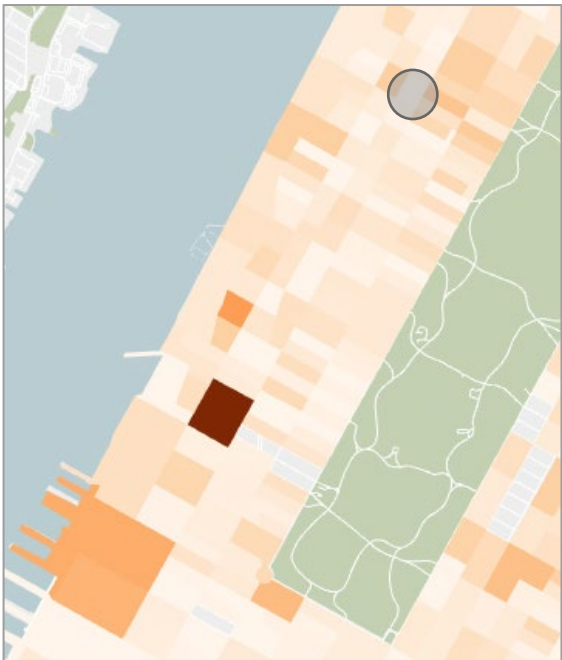


August



October

Accessibility: assessing hazards for older adults



SONYC: Sounds of New York City



- 55 sensors
- 3 boroughs
- Over 75 terabytes of audio data



Deployment numbers

55

Sensors deployed

1

Mobile sensor

5

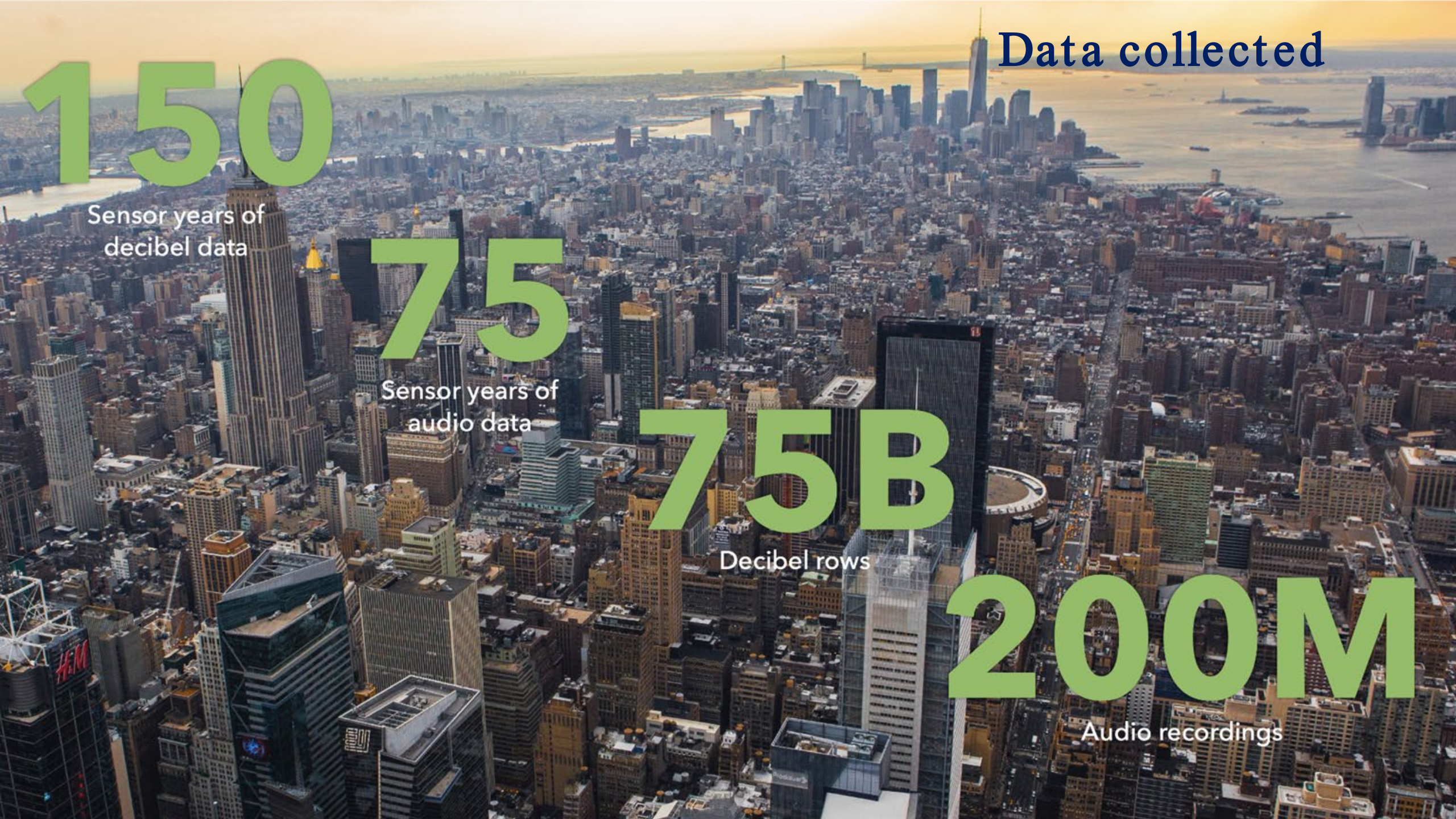
Years of sensing

7

City agencies

3

Business improvement
districts



Data collected

150

Sensor years of
decibel data

75

Sensor years of
audio data

75B

Decibel rows

200M

Audio recordings

SONYC

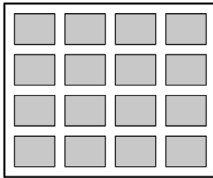


- Goal: improving quality of life through smart noise monitoring.
- Established sensor network and expertise:
 - Increased civic engagement in noise awareness.
 - Collaboration with key city agencies and infrastructure providers.
- Convergence research around data.

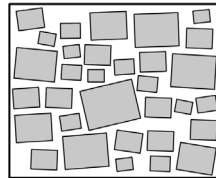
Urban data exploration

Image data

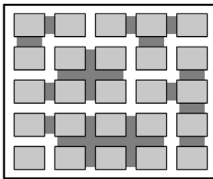
Grids



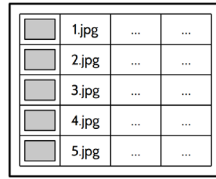
Collages



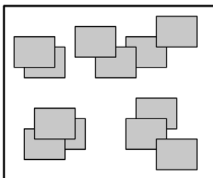
Similarity Grids



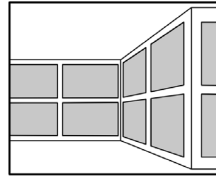
Spreadsheets



2D Mappings



3D Projections

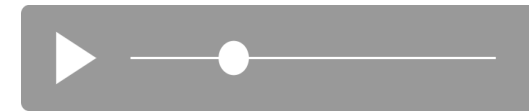


- Natural pictorial representation of image data
- Humans are able to visualize and understand sets of images in a *parallel approach*
- Organization approaches for images were proposed in the past. These approaches try to optimize the observation of specific **patterns present in collections of images**

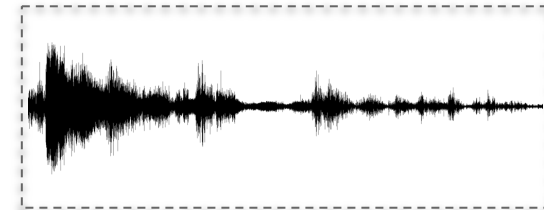
Urban audio data exploration

Audio data

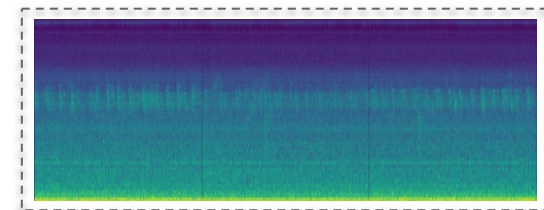
- Audio recordings are consumed in a **serial** way by us.
- To understand events happening in a 10-second audio snippet, users **must listen through the entire recording**.
- Although the **visualization of specific frequencies or loudness** can help identify interest periods of the recording, it is still difficult to build a **semantic understanding** of the recording.



Waveform



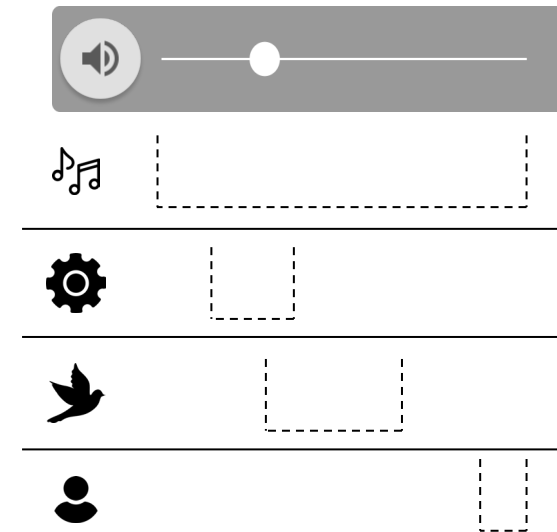
Spectrogram



Urban audio data exploration

Audio data

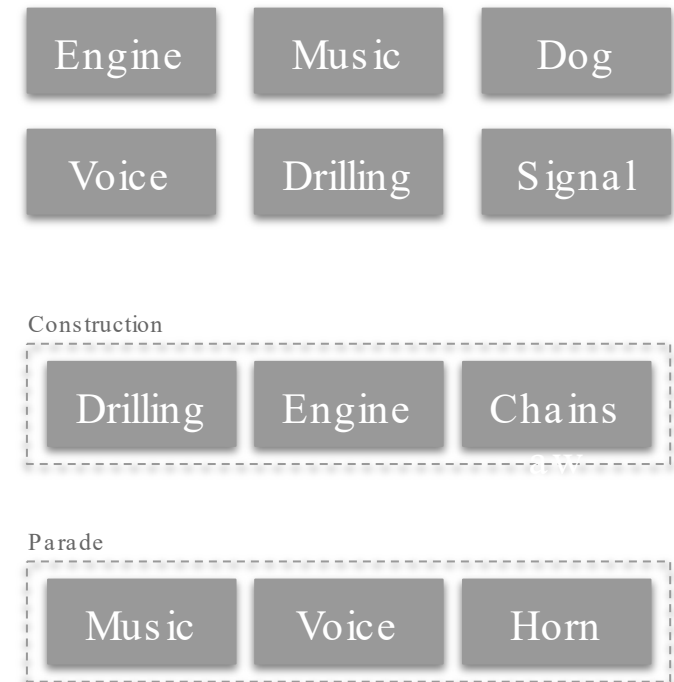
- Unlike images, where visual objects are opaque, sound objects are conceptually *transparent*, meaning that multiple objects (sound sources) can have energy at the same frequency.
- At any given instant in time, a sound recording might have a **mixture of background** (birds, dog barks) and **foreground sounds** (party, sirens).



Urban audio data exploration

Audio data

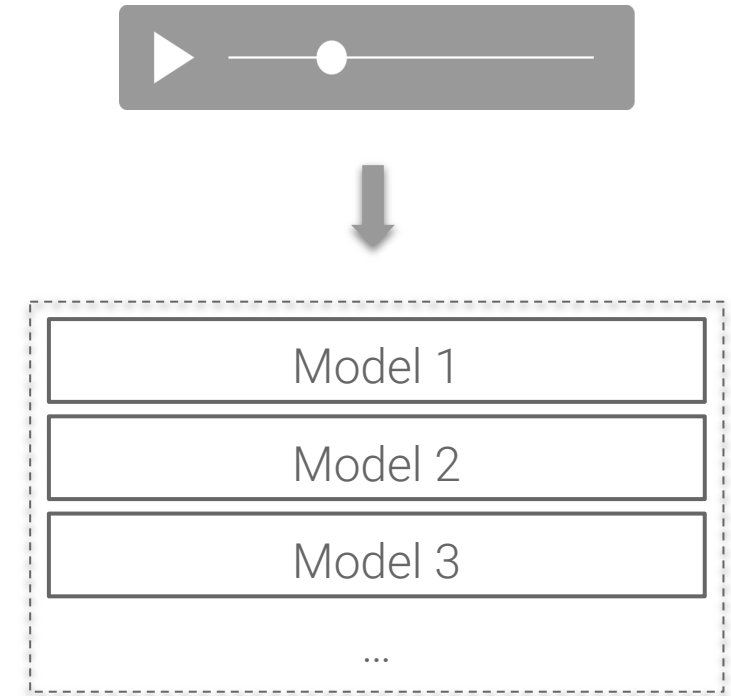
- **Class vocabularies are limited** , providing a **narrow view of the rich and varied soundscape** of the city
- Also, they do not account for classes that are a composition of previously created classes



Urban audio data exploration

Audio data

- The **scarcity of labeled urban audio data** makes it hard to generate models capable of transforming audio into representations that can represent different audio classes.
- Also, the **complexity of the urban soundscape** makes it even harder for models to be representative of such a dynamic environment.



Urban Rhapsody

[R1] **Interactive Identification and Labeling of Similar Sounds** : Given the highly complex acoustic environment we observe in cities, audio representations cannot clearly encode specific audio events that users might be interested in.

[R2] **Projection Steering Based on User Perception** : When exploring audio embeddings extracted from urban recordings through multidimensional projections, we often recognize clusters that do not represent the user's perception of the soundscape

[R3] **Iterative Creation of Classification Models** : Considering that current machine listening models present certain limitations, the system should provide the capability to iteratively create new classification models based on the data points labeled by the user (and, therefore, the user's perception of the soundscape)

Urban Rhapsody: Large-scale Visual Exploration of Urban Soundscapes, EuroVis 2022

Urban Rhapsody

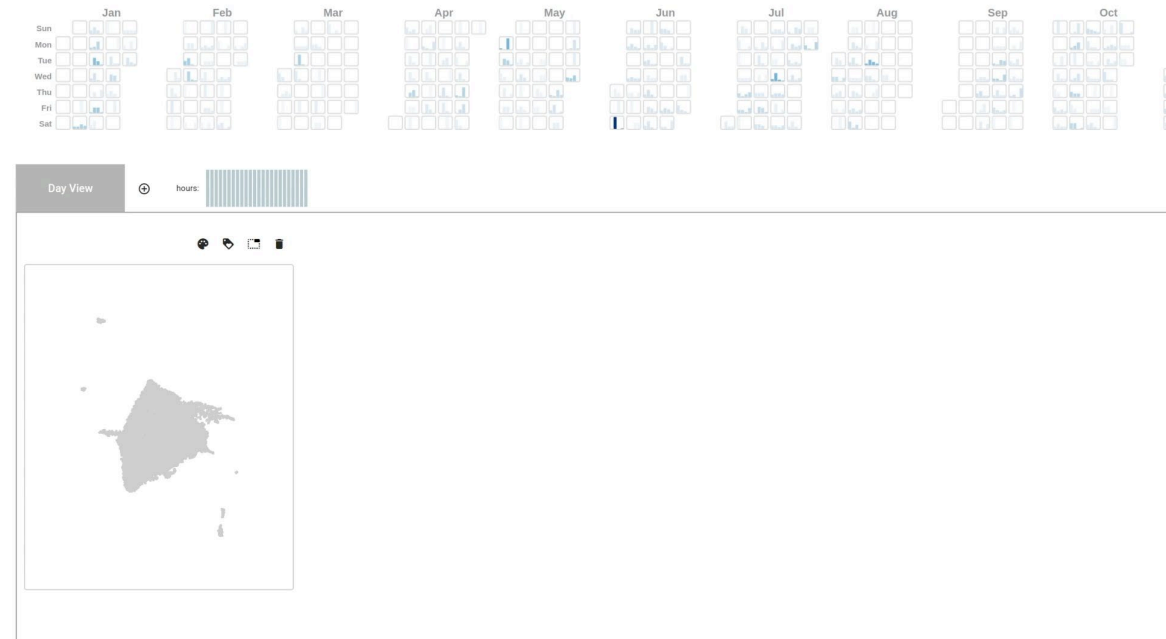


Urban Rhapsody: Creating prototypes

Binary Classification Model



Representatives



Bioacoustics monitoring

- Does exposure to loud urban noise lead to significant changes in bird's song traits and the time and frequency of their chorus?
- Do loud siren noises halt birds' dawn chorus?
- Are the birds nesting in noisy urban areas like Manhattan local parks adapted to the level of urban noise?

Bioacoustic monitoring

